

遥操&无本体数据 实验记录

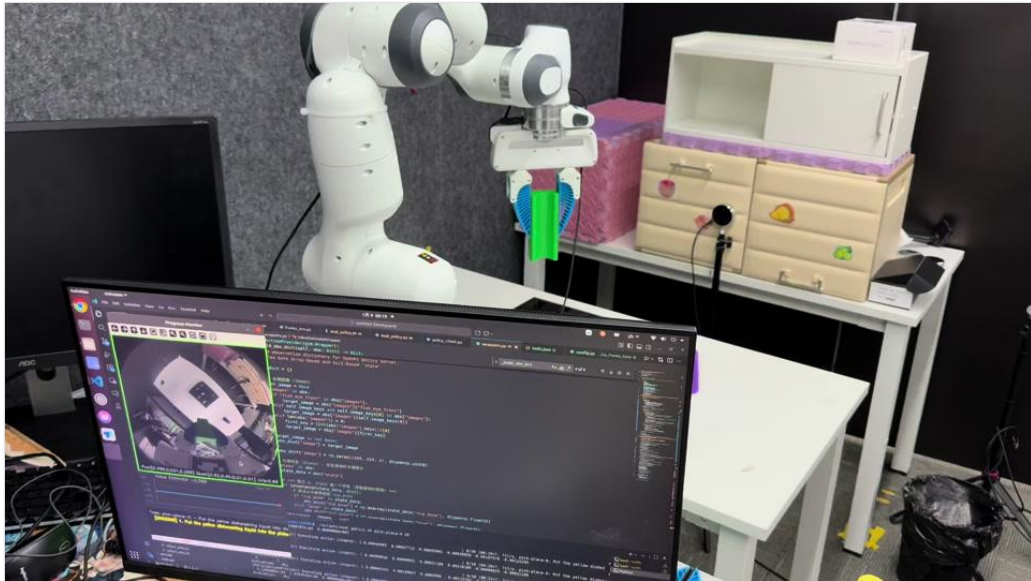
任务：拿起杯子放盘子里

【遥操数据】 pick&place cup

第一阶段 输入图像+输出delta action

franka_place_0107 固定UMI夹爪

- 数据路径：



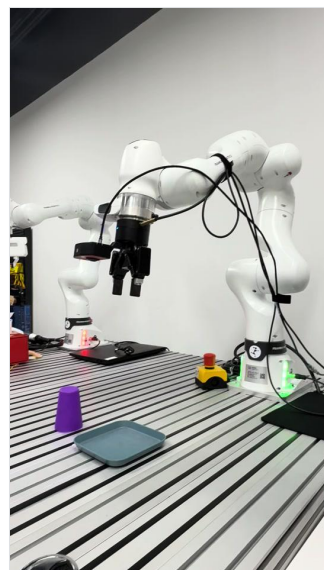
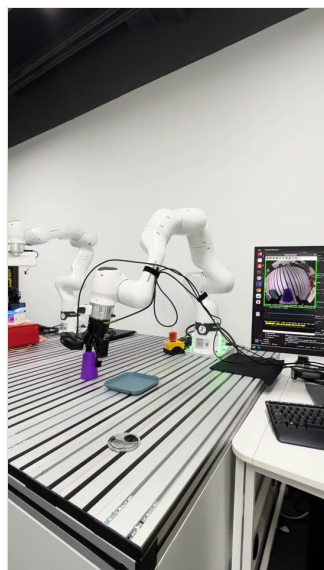
franka_pick_place_0112 setup搜狐franka

- ckpt路径： /home/ubuntu/qiuyi/ckpts/ckpt_franka_pick_place_0113



franka_pick_place_0115 调整夹爪坐标系+采20条

- ckpt路径: /home/ubuntu/qiuyi/ckpts/ckpt_franka_pick_place_0116
- 效果: 成功率比较低



franka_pick_place_0118 采100条

- ckpt路径: /home/ubuntu/qiuyi/ckpts/ckpt_franka_pick_place_0119/10000
- 效果: 位置泛化性明显提升，但是成功率并不高

demo

 **IMG_3014.MOV**
202.33MB

 **IMG_3013.MOV**
206.21MB

成功率

左侧					右侧				
1	2	3	4	5	6	7	8	9	10
3/5	2/5	1/5	2/5	2/5	0	0	0	0	1
上方					下方				
11	12	13	14	15	16	17	18	19	20
1	1	0	0	1	0	1	0	0	0

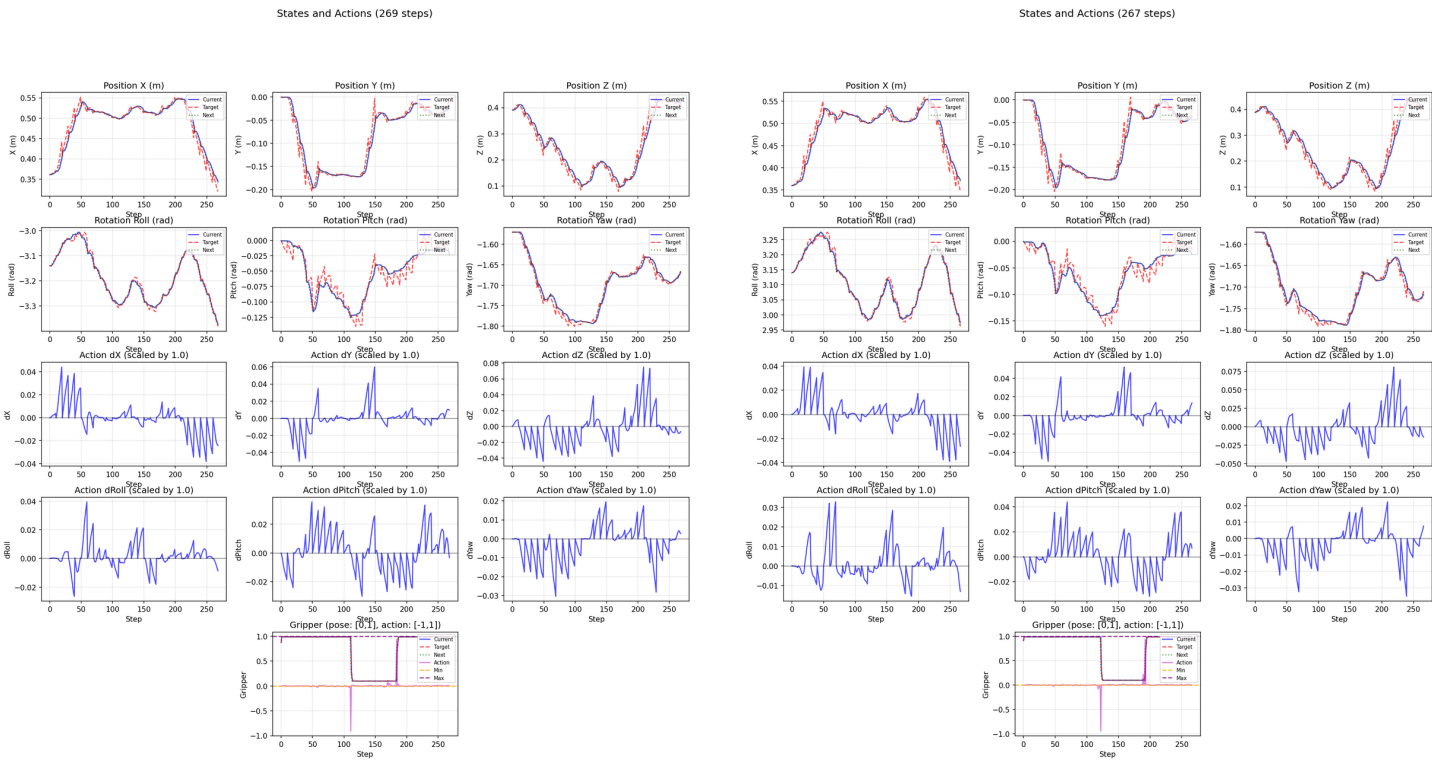
franka_pick_place_0121 140条 基础+干扰+retry数据

成功率: 1.5w

左侧（100% 四次retry）					右侧（60% 数据少）				
1	2	3	4	5	6	7	8	9	10
1	1	1	1	1	1（五进宫）	1（七进宫）	1（重下）	0	0
上方（100% 一次retry）					下方（60%）				
11	12	13	14	15	16	17	18	19	20
1	1	1	1	1	0	1	1	1	0

第二阶段 输入图像+输出relative action

franka_pick_place_0118 采100条

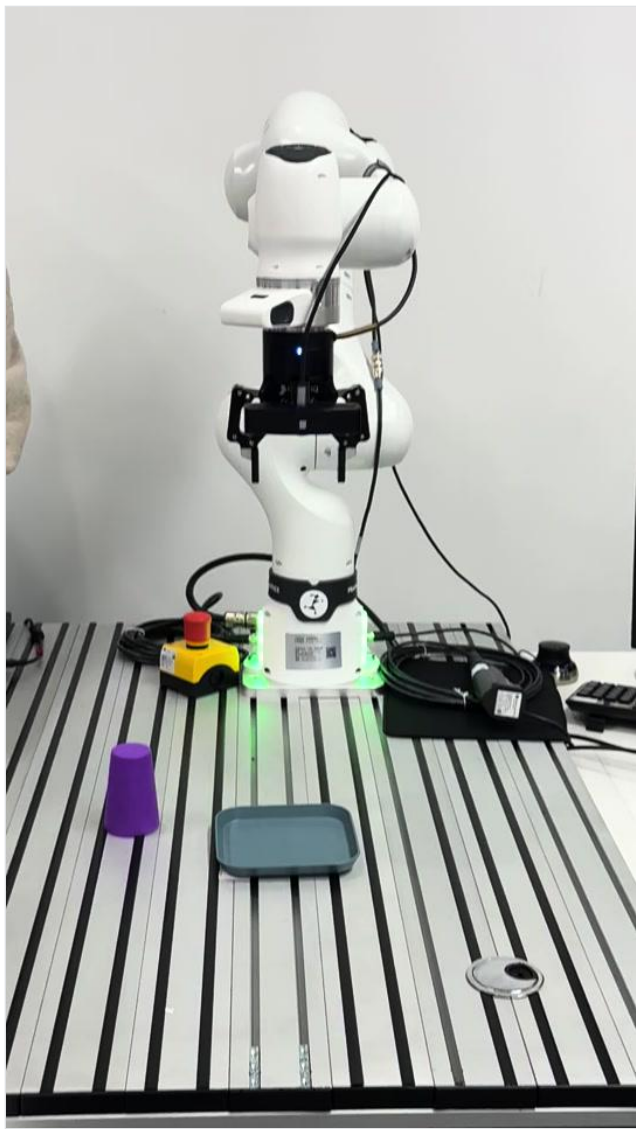


franka_pick_place_0121 140条 基础+干扰+retry数据

成功率：1.5w

左侧（100% 全部一次成功！！）					右侧（0% 数据少）				
1	2	3	4	5	6	7	8	9	10
1	1	1	1	1	0	0	0	0	0
上方（20%）					下方（100% 全部成功）				
11	12	13	14	15	16	17	18	19	20
0	1	0	0	0	1	1	1	1	1

demo：1.5w左，2w右



第三阶段挂起 输入图像&relative state + 输出relative action

任务比较简单，不需要输入state

【无本体数据】 pick&place cup

0104_100_test

现在采集了128条，留28条作为验证集，使用前100条训练

处理之后发现数据似乎有问题，采的时候每次停止位置不太对

```

1 {"episode_index": 0, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 501}
2 {"episode_index": 1, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 280}
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4 {"episode_index": 3, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 419}
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15 {"episode_index": 14, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 335}
16 {"episode_index": 15, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 263}
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21 {"episode_index": 20, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 65}
22 {"episode_index": 21, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 146}
23 {"episode_index": 22, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 119}
24 {"episode_index": 23, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 70}
25 {"episode_index": 24, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 62}
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32 {"episode_index": 31, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 109}
33 {"episode_index": 32, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 59}
34 {"episode_index": 33, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 61}
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37 {"episode_index": 36, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 67}
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39 {"episode_index": 38, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 74}
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42 {"episode_index": 41, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 484}
43 {"episode_index": 42, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 427}
44 {"episode_index": 43, "tasks": ["grasp the yellow dishwashing liquid bottle and place it on the scale."], "length": 370}

```

数据的问题是过热引起的，同样是6s的视频，image只有70+个

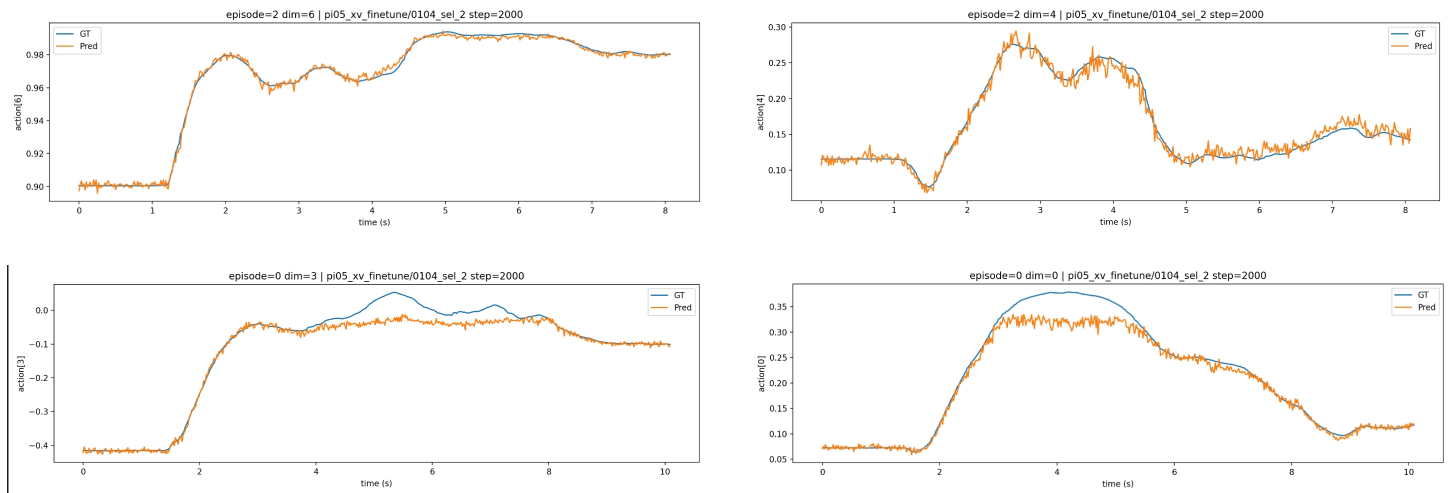
```

topics:      /xv_sdk/PoseStampedConfidence [3e0801c3f083917c5b04ba335d08b004] 1 msg : tf2_msgs/TFMessage
              /tf_static
              /xv_sdk/250801DR48FP25002565/clamp/Data 266 msgs : xv_sdk/Clamp
              /xv_sdk/250801DR48FP25002565/color_camera/camera_info 79 msgs : sensor_msgs/CameraInfo
              /xv_sdk/250801DR48FP25002565/color_camera/image 79 msgs : sensor_msgs/Image
              /xv_sdk/250801DR48FP25002565/slam/pose 638 msgs : xv_sdk/PoseStampedConfidence
              /xv_sdk/250801DR48FP25002565/slam/pose_markers 12 msgs : visualization_msgs/MarkerArray
(base) nics@nics-X299-UD4-EX:~/CSW/openpi/umi/data_0104_val$ rosbag play 2026-01-04-16-34-39.bag -r 0.1
[ERROR] [1767775946.258396852]: [registerPublisher] Failed to contact master at [localhost:11311]. Retrying...
^C(base) nics@nics-X299-UD4-EX:~/CSW/openpi/umi/data_0104_val$ rosbag play 2026-01-04-16-34-39.bag -r 0.1
[ INFO] [1767775965.569988304]: Opening 2026-01-04-16-34-39.bag
Waiting 0.2 seconds after advertising topics... done.
Hit space to toggle paused, or 's' to step.
[RUNNING] Bag Time: 1767515680.561306 Duration: 0.585405 / 6.760208 5.84
(base) nics@nics-X299-UD4-EX:~/CSW/openpi/umi/data_0104_val$ rosbag play 2026-01-04-16-34-39.bag
[ INFO] [1767775975.624317134]: Opening 2026-01-04-16-34-39.bag
Waiting 0.2 seconds after advertising topics... done.
Hit space to toggle paused, or 's' to step.
[RUNNING] Bag Time: 1767515686.733230 Duration: 6.757329 / 6.760208 5.88

```

对齐是固定按60Hz对齐，导致出来的视频也差了很多

跑4k步挂掉了，需要查bug



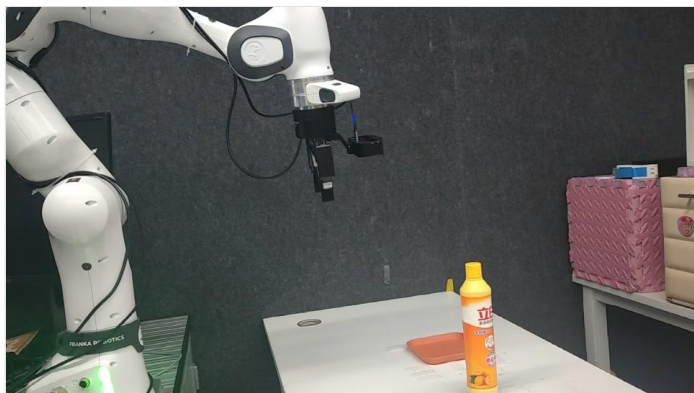
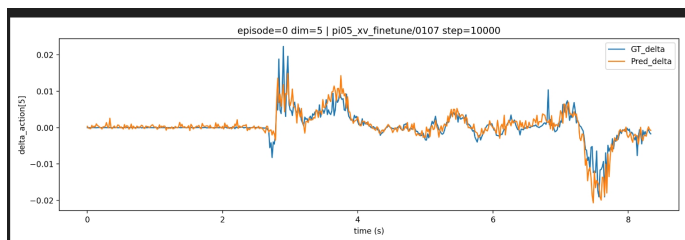
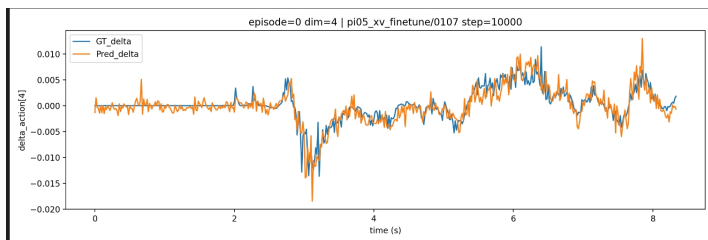
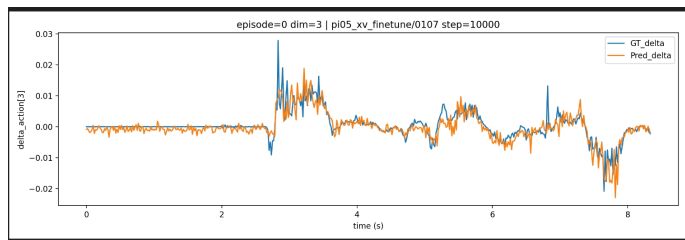
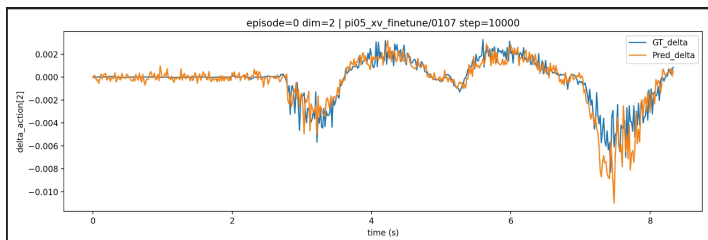
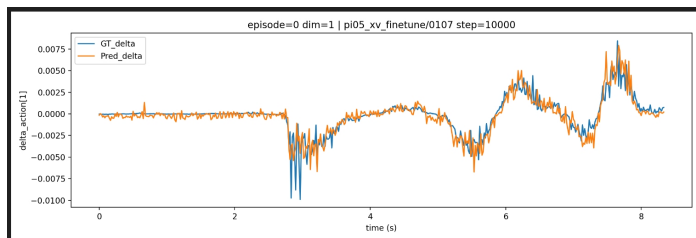
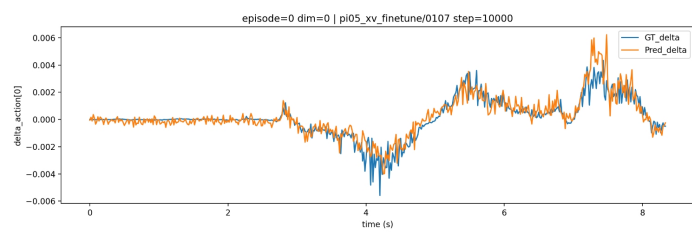
0107

delta state, delta action, 夹爪不delta, 按照帧delta

95条数据，一部分绿盘子，一部分红盘子

```
lr_schedule=_optimizer.CosineDecaySchedule(  
    warmup_steps=1_000,  
    peak_lr=5e-5,  
    decay_steps=1_000_000,  
    decay_lr=5e-5,  
)  
log_interval=500,  
save_interval=2000,  
keep_period=10_000,  
num_workers=1,  
num_train_steps=30_000,  
batch_size=16,
```

1w个step下，感觉delta有点太细了



0114

不需要对齐，就使用relative pose

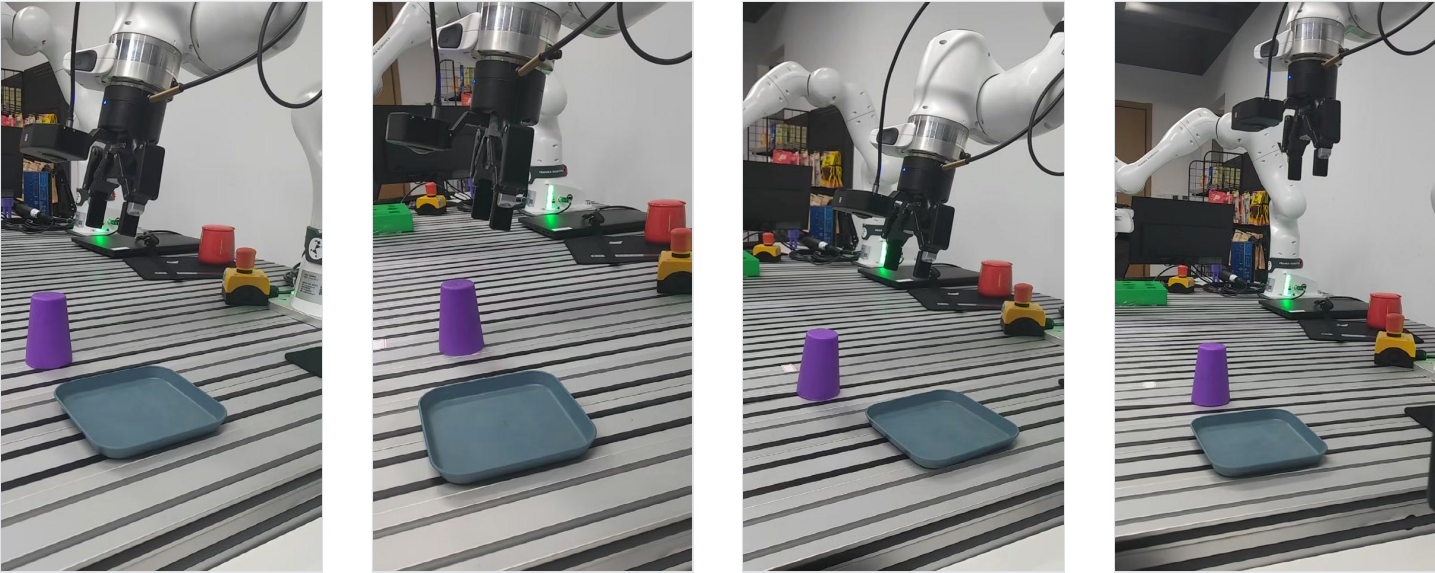
输入只是图像，输出是相对于chunk第一帧的relative

部署效果很差，只是向下走

0116：灵智初开

输入只有图像，训练数据downsample 到 10Hz，action chunk = 16

action 11 维度： pose + 6drot + 1 + gripper
泛化性很差，杯子位置和初始位置都需要调一调



遥操无本体抓杯子成功率对比

setting 1： 杯子在盘子一侧

setting 2： 杯子在盘子旁任意位置

pos1: 遥操默认初始位置

pos2: 遥操默认初始位置往下一点，夹爪有所移动

无本体 40 1 1

0	0	0	0	0	0	0	1	1	0
0	0	0	0	0	0	0	0	0	0

无本体 40 1 2

0	0	1	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	0

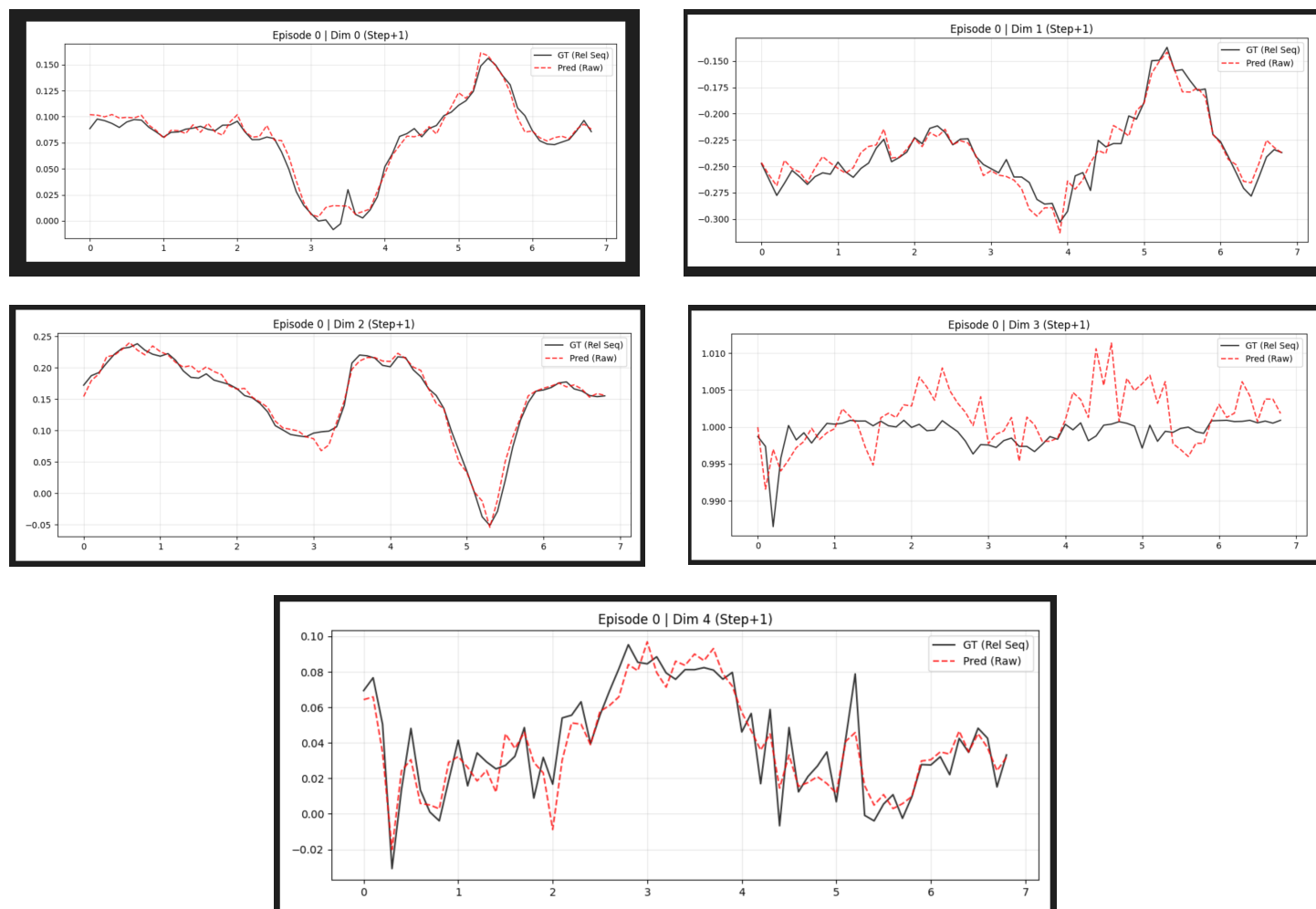
200条数据：发现后面160条gripper维度有问题，都在73附近飘

action 10维度：pose + 6drot + gripper

gripper[0,88], output时 /88 赋 -1 或者 1

效果比较差，只会往前夹

offline结果：



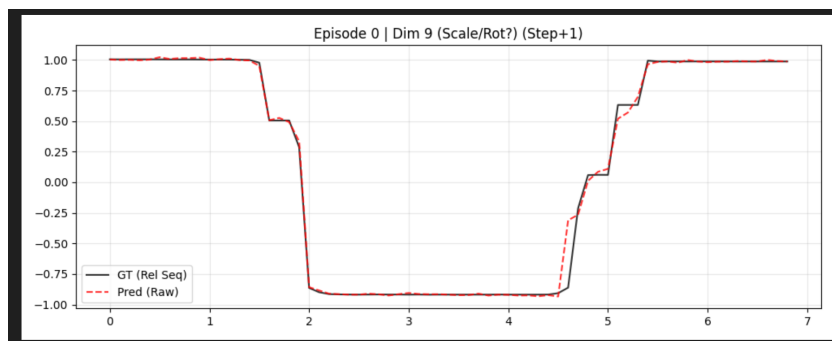
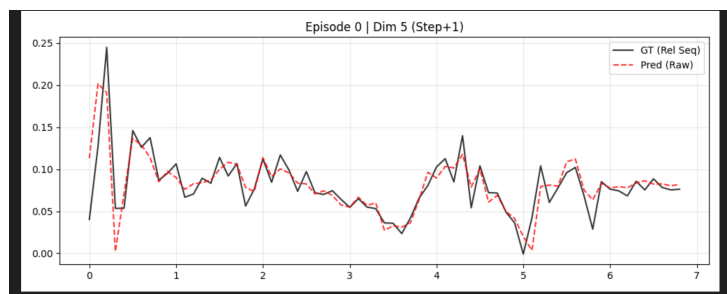
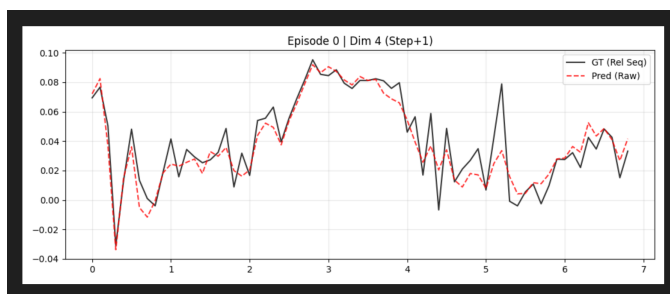
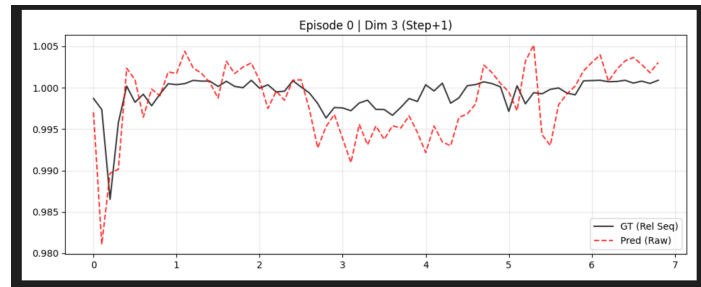
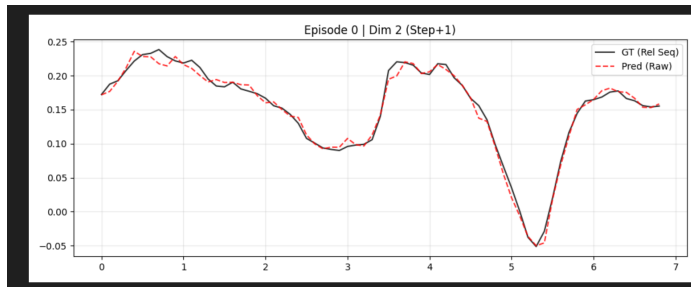
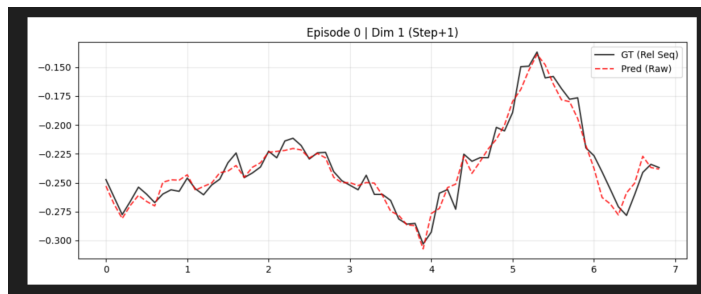
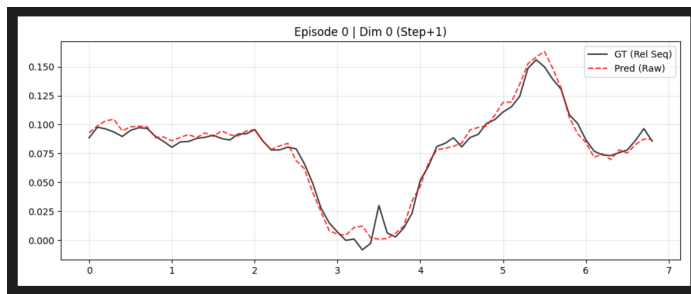
0120

也是数据本身有问题

batch_size调大至128，

取训练1w step结果

offline



0122

200条数据：retry + 普通

lerobot的action 存下一个时刻的状态，以此解决每个chunk第一帧是 0 的问题

setting 1： batch_size = 16， downsample 到 10 Hz， 原始方法进行norm

setting 2： batch_size = 16， downsample 到 20 Hz， 原始方法进行norm

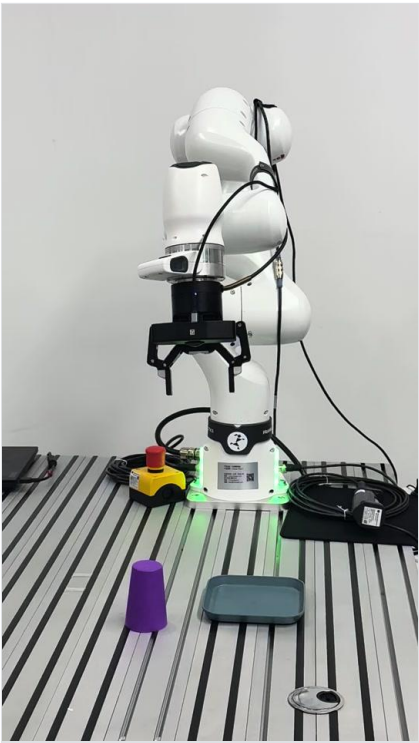
setting 3： batch_size = 16， downsample 到 10 Hz， 新方案进行norm， 每一个T分别做norm

setting 4： batch_size = 16， downsample 到 20 Hz， 新方案进行norm， 每个T分别做norm

成功率：s1模型 1w

左侧（80%）						右侧（58%）				
1	2	3	4	5		6	7	8	9	10
1	1	1	0	1		1	1	0	0	0.9（杯子在盘边缘）
上方（40% 数据少）						下方（98% 习惯retry一次）				
11	12	13	14	15		16	17	18	19	20
0.5（夹住了倾倒的杯子）	0	0.5（夹住了杯子又松开了）	1	0		1	1	1	1	0.9（杯子在盘边缘）

demo：s1 1w



成功率：s2 1w

左侧						右侧（58%）				
1	2	3	4	5		6	7	8	9	10
观感不如0122-s1，夹不到						ok了				
上方（40% 数据少）						下方				
11	12	13	14	15		16	17	18	19	20

【takeaway】 1.12-1.23

todo

1. 数据采集方面

- 每批数据都需要抽查做replay，或者隔一段时间抽查看state分布情况
- retry和干扰数据很重要，但是比例不能太高（从失败恢复很重要，进入失败可以删掉）
- 每次采集指定一块区域，不要是整个工作台
- 采集设备末端坐标系要和部署设备末端执行器的坐标系一致（setting里改）
- 后续采集增加diverse背景
- 鱼眼相机去畸变

2. 数据处理方面

- 图像的输入频率很重要--> franka 10Hz好使
- robotiq的gripper action用binary模式，改continuous(针对易碎易形变物品)
- 上传服务器的时间后续优化减少
- 统一现在无本体和遥操数据处理接口的脚本
- 支持lerobot格式数据集的增量式扩充

3. 算法方面

- 归一化时忽略gripper那一维
- relative在于somehow消除掉base坐标系
- step 1w和2w区别不大
- 初始loss得在2以下，如果很大说明代码有问题
- chunk分时间步归一化
- 模型增加state输入（提高优先级），主要看双臂怎么
- 试下cotrain

4. 评测方面

- loss其实没啥用，只能看到输入输出有没有对齐，还得看offline的测试结果
 - offline时旋转表征拟合较差，想想咋回事
- target和实际位置可能因控制器有区别，可以画图看一下（jet）

- c. 怎么快速评测很多setting下训出来的ckpt
 - d. 解决部署时franka执行的卡顿问题
5. 科研idea方面
- a. 分工调研，思考可能的创新点，target CoRL'26

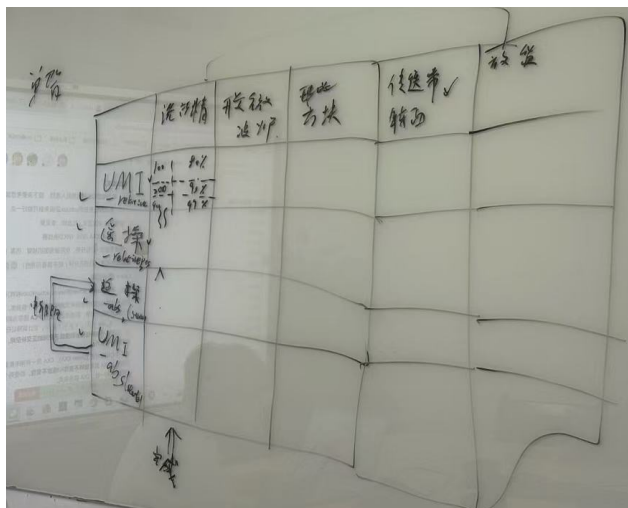
【TODO】 1.24

1. 单手方块放到柜子任务看是不是work 陈帅文 秋怡 顾秋怡

- a. 数据给楚风：遥操无本体都行

2. 数据效率表格 秋怡 顾秋怡 小龙 单小龙

- a. 加一个setting，UMI数据，使用相对于第一帧的action
- b. 测一下pi05本身能力



3. 双手hand over：递瓶子

- a. 无state，图像隐式来做
 - i. 看好不好使
 - ii. 好使之后做一些消融，让他彼此看不到，inpainting到背景
- b. 虚拟位姿给绝对state，绝对pose，target Franka 朝阳 申朝阳
 - i. 一条轨迹给多个虚拟base和位姿——>数据增强
- c. umi relative state输入

4. 采数据再采一个第三视角的图像 罗兆亿

5. 触觉方式给 MS --> M

任务：方块先后放抽屉

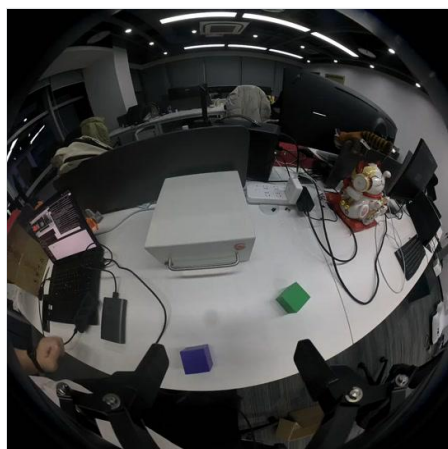
【无本体数据】

0123

3个背景, 35条数据, rosbag 130G——>有些pose会录丢, 也需要筛出去

```
18 :eef_pos": {"min": [-0.176876500248909, -0.1794065535068512, -0.30686476826667786], "max": [0.0853232592344284, 0.1152
19 :f_pos": {"min": [-0.17229372262954712, -0.1590307205915451, -0.22038917243480682], "max": [0.03688020631670952, 0.12
20 'eef_pos": {"min": [0.0, 0.0, 0.0], "max": [0.0, 0.0, 0.0], "mean": [0.0, 0.0, 0.0], "std": [0.0, 0.0, 0.0], "count":
21 "eef_pos": {"min": [0.0, 0.0, 0.0], "max": [0.0, 0.0, 0.0], "mean": [0.0, 0.0, 0.0], "std": [0.0, 0.0, 0.0], "count":
22 "eef_pos": {"min": [0.0, 0.0, 0.0], "max": [0.0, 0.0, 0.0], "mean": [0.0, 0.0, 0.0], "std": [0.0, 0.0, 0.0], "count":
23 :eef_pos": {"min": [0.0, 0.0, 0.0], "max": [0.0, 0.0, 0.0], "mean": [0.0, 0.0, 0.0], "std": [0.0, 0.0, 0.0], "count":
24 'eef_pos": {"min": [0.0, 0.0, 0.0], "max": [0.0, 0.0, 0.0], "mean": [0.0, 0.0, 0.0], "std": [0.0, 0.0, 0.0], "count":
25 "eef_pos": {"min": [NaN, NaN, NaN], "max": [NaN, NaN, NaN], "mean": [NaN, NaN, NaN], "std": [NaN, NaN, NaN], "count":
26 "eef_pos": {"min": [NaN, NaN, NaN], "max": [NaN, NaN, NaN], "mean": [NaN, NaN, NaN], "std": [NaN, NaN, NaN], "count":
27 :eef_pos": {"min": [NaN, NaN, NaN], "max": [NaN, NaN, NaN], "mean": [NaN, NaN, NaN], "std": [NaN, NaN, NaN], "count":
28 :f_pos": {"min": [-0.18221727013587952, -0.19946220517158508, -0.5857147574424744], "max": [0.07122945785522461, 0.12
29 , "eef_pos": {"min": [-0.07789292186498642, -0.23016802966594696, -0.2708224952220917], "max": [0.30680668354034424,
30 :eef_pos": {"min": [-0.14436577260494232, -0.2639302909374237, -0.23891299962997437], "max": [0.07071684300899506, 0.0
31 'eef_pos": {"min": [-0.08722511678934097, -0.2177821397781372, -0.21013763546943665], "max": [0.2801908254623413, 0.0
32 :eef_pos": {"min": [-0.16135373711585999, -0.18644262850284576, -0.18141129612922668], "max": [0.0820620134472847, 0.0
```

数据示例:



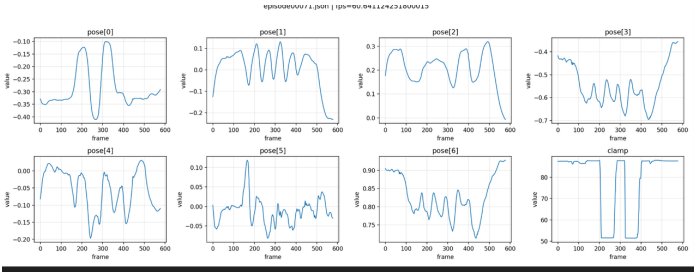
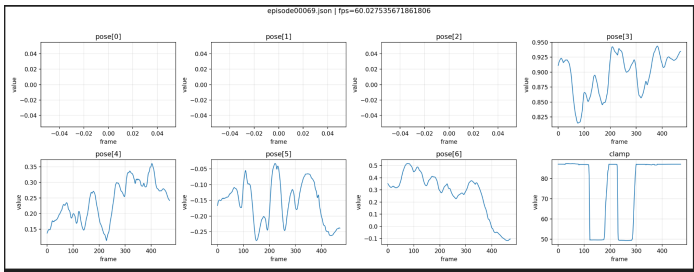
0125

76条数据, 249G

改变开抽屉方式

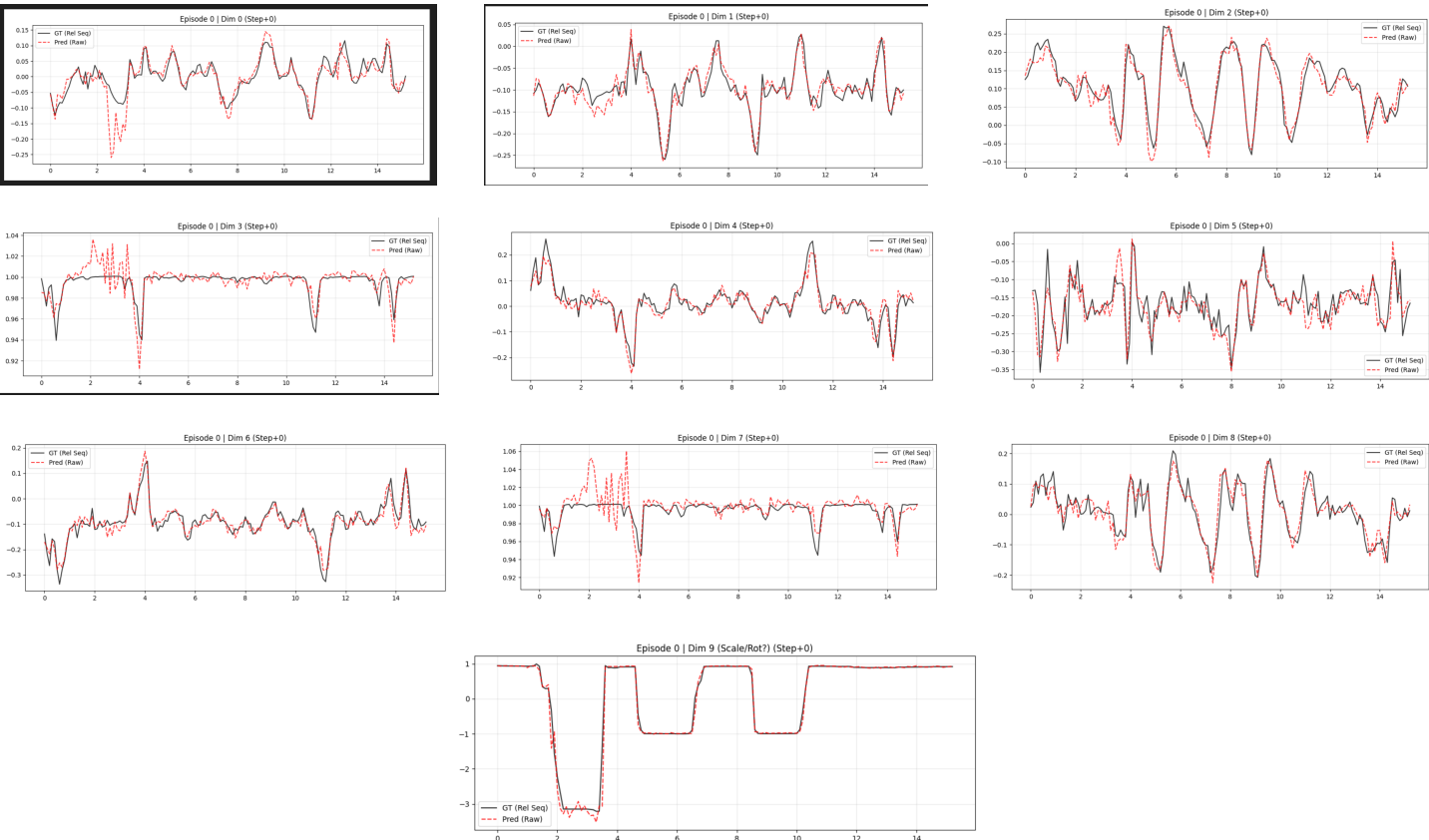


做出state分布情况, 筛数据

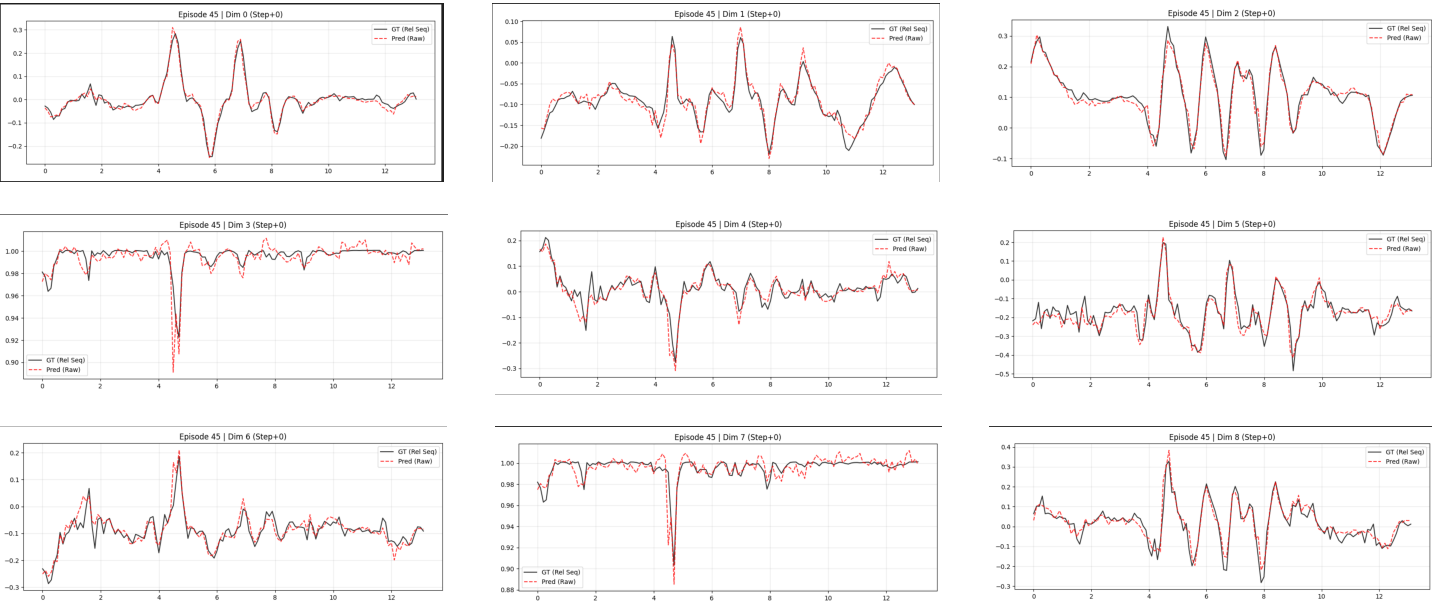


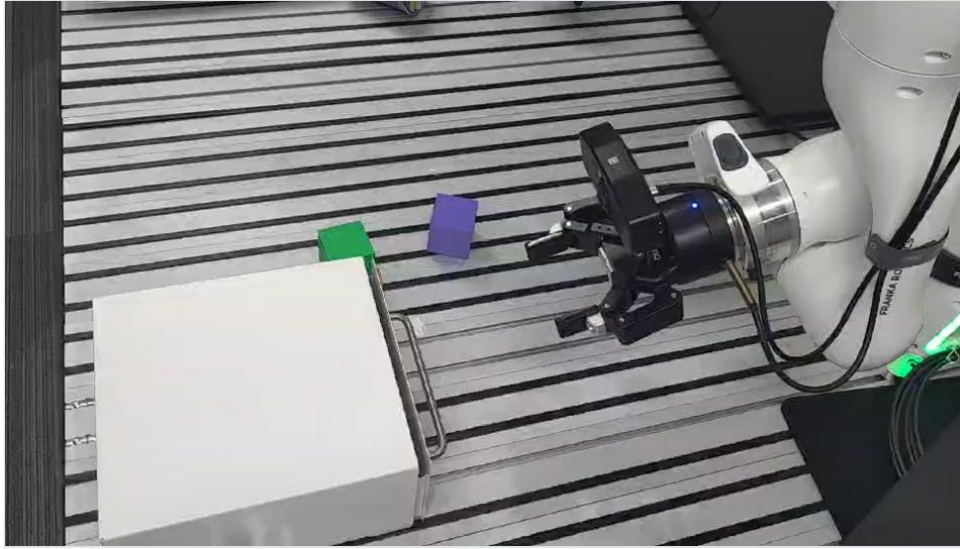
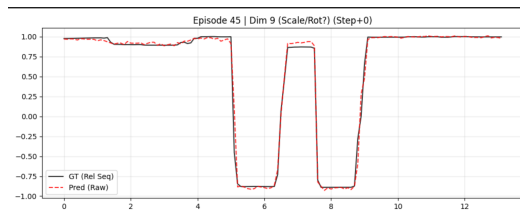
offline测试情况：

第一种开抽屉方式：



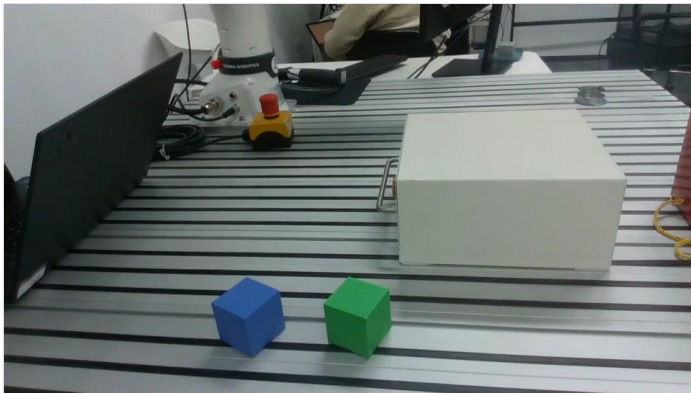
第二种开抽屉方式：





0127_w

加第三视角（D435），85条数据



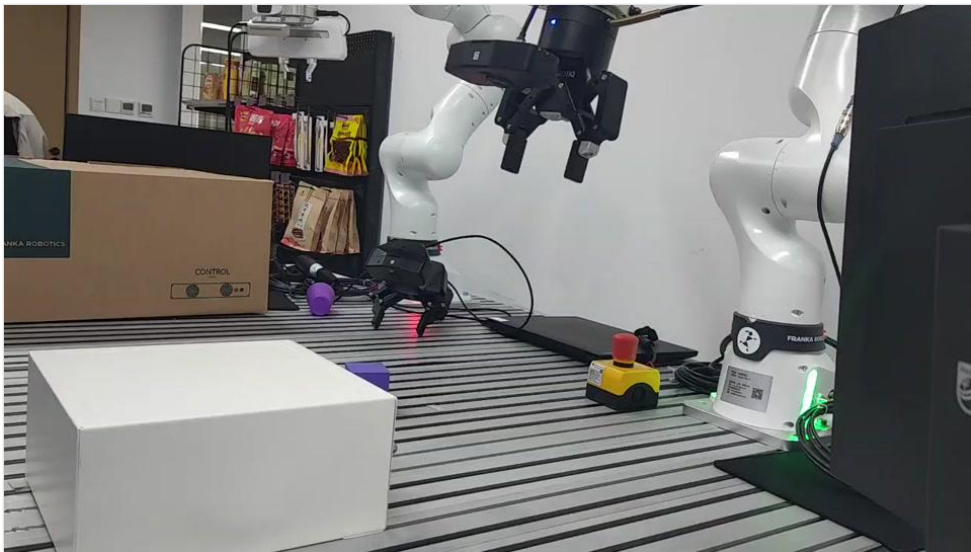
修改较多，在fastumi文档新开了内容

0.127_wo

将现在的数据与0125的数据直接合并，158条数据来训

【少量遥操数据】

0125



【TODO】：

1. 考虑子任务划分
2. 鱼眼相机去畸变

Previous pipeline

数采+pi0.5微调+部署

step1-1 遥操数据采集

- `demo_data` 文件夹下Parquet格式的文件

4090台式机

```
1  bash ./run_server.sh # 启动realRL的robot server
2
3  CYCLIC=false ./scripts/collect_data.sh {task_name} 100 #3D鼠标采集操作数据
```

step2-1 遥操数据处理

4090台式机

```
1  ./scripts/process_data_local.sh {task_name} #将数据转成mp4等格式
2
3  scp -r ready_to_upload/{task_name} eva14:/share/guqiuyi-
   local/franka_ready_to_upload # 上传服务器
```


eva14服务器

```
1 INPUT_DIR=/share/guqiuyi-local/franka_ready_to_upload/{task_name} \  
2 REPO_ID=local/franka_pick_place_cubes_0125 \  
3 ./scripts/process_data_server.sh #将数据转成lerobot格式
```

step1-2 无本体数据采集

笔记本/4090台式机

```
1 #cd ~/catkin_ws/  
2 roslaunch xv_sdk xv_sdk.launch  
3  
4 #cd ~/qiuyi/FastUMI/FastUMI_Monitor_Tool/  
5 cd ~/CSW/FastUMI_Monitor_Tool/  
6 bash fastumi_monitor_menu.sh # 按20和21查看图像和夹爪数值  
7  
8 cd ~/CSW/data  
9 rosbag record \  
10 /xv_sdk/250801DR48FP25002268/clamp/Data \  
11 /xv_sdk/250801DR48FP25002268/color_camera/image \  
12 /xv_sdk/250801DR48FP25002268/slam/pose # 注意序列号
```

step2-2 无本体数据处理

笔记本/台式机

```
1 # 将rosbag批量处理成mp4和json (convert_rosbag_to_mp4_vis.py)  
2 ./batch_stage1_vis.sh  
3  
4 # 根据可视化图像删去有问题的数据  
5  
6 # 删完重新排序  
7 python json_sort.py --dir xxx  
8 python json_sort.py --dir xxx --apply  
9  
10 # 压缩文件并传输  
11 scp data.zip eva14:/share/guqiuyi-local/umi_ready_to_upload
```

eva14服务器

```
1 # 将mp4和json处理成lerobot格式  
2 unzip data.zip
```

```

3
4 export HF_LEROBOT_HOME=/share/guqiuyi-local/huggingface/lerobot
5
6 python convert_mp4_data_to_lerobot_downsample.py \
7 --stage1_dir=/share/guqiuyi-local/umi_ready_to_upload/batch_stock_shelves_0128
8 \
9 --repo=local/{task_name}
10
11 python merge_and_sample_lerobot.py --root /share/guqiuyi-
local/huggingface/lerobot --sources local/umi_pick_place_cup_0121-
local/umi_pick_place_cup_0130 --base_output_name-
local/umi_pick_place_cup_0130_eval --counts 100 200 400
12
13 python convert_mp4_to_lerobot_sampling.py \
14 --input_dirs /share/guqiuyi-
15 local/umi_readto_upload/batch_pickup_cup_0121_200episode /share/guqiuyi-
16 local/umi_ready_to_upload/batch_pickup_cup_0130_224episode \
17 --repo local/umi_pick_place_cup_0130_eval100 \
18 --num_episodes 100 \
19 --seed 42
20
21 python convert_mp4_to_lerobot_sampling.py \
22 --input_dirs /share/guqiuyi-
23 local/umi_readto_upload/batch_pickup_cup_0121_200episode /share/guqiuyi-
24 local/umi_ready_to_upload/batch_pickup_cup_0130_224episode \
25 --repo local/umi_pick_place_cup_0130_eval200 \
26 --num_episodes 200 \
27 --seed 42
28
29 python convert_mp4_to_lerobot_sampling.py \
30 --input_dirs /share/guqiuyi-
31 local/umi_readto_upload/batch_pickup_cup_0121_200episode /share/guqiuyi-
32 local/umi_ready_to_upload/batch_pickup_cup_0130_224episode \
33 --repo local/umi_pick_place_cup_0130_eval400 \
34 --num_episodes 400 \
35 --seed 42

```

目前还在debug:

4090台式机

```

1 process_umi_data_local.sh #将数据转成mp4等格式
2
3 scp /home/xxx.zip eva14:/share/guqiuyi-local/umi_ready_to_upload # 上传服务器

```

eval14服务器

```
1  unzip xxx.zip
2
3  export HF_LEROBOT_HOME=/share/guqiuyi-local/huggingface/lerobot
4
5  uv run umi_scripts/process_umi_data_server.py \
6      --input_dir="/share/guqiuyi-
7      local/umi_ready_to_upload/batch_pickup_cubes_0125" \
8      --repo_id="local/umi_pick_place_cubes_0128" \
9      --target_fps=10 \
10     --task_text="pick and place cubes"
```

step3 模型微调

- 改 `src/openpi/training/config.py` 中的 `repo_id="local/{task_name}"`

eval14服务器

```
1  export HF_LEROBOT_HOME=/share/guqiuyi-local/huggingface/lerobot # LeRobot 本地数
   据根目录
2
3  CUDA_VISIBLE_DEVICES=0 uv run scripts/compute_norm_stats.py --config-name
   pi05_xv_finetune # 归一化
4
5  CUDA_VISIBLE_DEVICES=0 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
   scripts/train.py pi05_xv_finetune --exp-name=ckpt_umi_stock_shelves_0129 --
   overwrite # 开始训练
6
7
8  CUDA_VISIBLE_DEVICES=3 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
   scripts/train.py pi05_umi_finetune --exp-
   name=ckpt_umi_pick_place_cup_0130_eval400_image --overwrite
9
10
11
12  CUDA_VISIBLE_DEVICES=1 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
   scripts/train.py pi05_xv_finetune_fast --exp-
   name=ckpt_umi_pickup_cup_0130_fast --overwrite # 开始训练
```

step4 策略部署

4090台式机

```
1  scp -r eval14:/share/guqiuyi-
```

```

1 local/checkpoints/pi05_franka_finetune/0113_test_pick_place
  /home/ubuntu/qiuyi/ckpts # 将模型复制到本地4090
2
3 bash ./run_server.sh # 启动realRL的robot server
4
5 uv run scripts/serve_policy.py policy:checkpoint --
  policy.config=pi05_franka_finetune --
  policy.dir=/home/ubuntu/qiuyi/ckpts/ckpt_franka_pick_place_0122/15000 # 启动
  openpi server
6
7 CYCLIC=false ./scripts/eval_policy.sh pick-place-0 20 # 启动realRL client

```

多任务评测

1.25-1.26

分两步处理遥操数据，缩短上传时间

`./scripts/process_data.sh {task_name}` 改成：

代码块

```

1 # 4090台式机
2 chmod +x scripts/process_data_local.sh
3 ./scripts/process_data_local.sh pick-place-
  0
4
5 # 传输
6 scp -r ready_to_upload
  username@eva14:/path/to/openpi/
7
8 # 服务器
9 chmod +x scripts/process_data_server.sh
10 INPUT_DIR=/share/guqiuyi-
  local/franka_lerobot_dataset/local/pick-
  place-cubes-0
  REPO_ID=local/franka_pick_place_cubes_0125_
  v1 ./scripts/process_data_server.sh

```

```

v ready_to_upload/pick-place-0
v plots
  pick-place-0_ep0000_filtered_states.png
  pick-place-0_ep0001_filtered_states.png
  pick-place-0_ep0000_filtered_fish_eye_front.mp4
  pick-place-0_ep0000_filtered_meta.json
  pick-place-0_ep0000_filtered.parquet
  pick-place-0_ep0001_filtered_fish_eye_front.mp4
  pick-place-0_ep0001_filtered_meta.json
  pick-place-0_ep0001_filtered.parquet

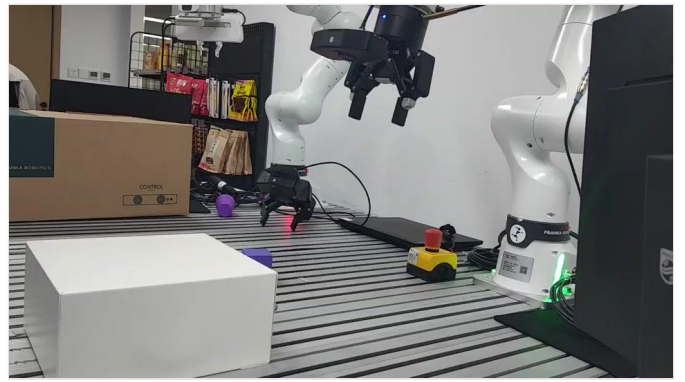
```

上传服务器的数据内容

一步处理	用delta	ckpt_franka_pick_place_cube_0125
	用relative	ckpt_franka_rel_pick_place_ubes_0125

两步处理	用delta	ckpt_franka_pick_place_cube_0125_v1
	用relative	ckpt_franka_rel_pick_place_ubes_0125_v1

部署测试了下没问题，后面就用这个方案



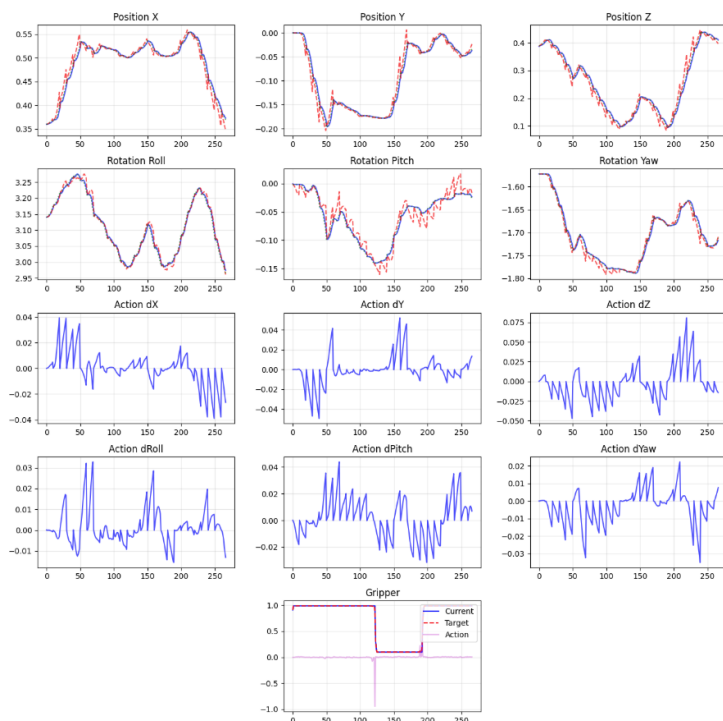
umi数据增加可视化

【pipeline】umi数据处理

```

1  # 本地：将rosvbag批量处理成mp4和json (convert_rosvbag_to_mp4_vis.py)
2  ./batch_stage1_vis.sh
3
4  # 传输
5  scp /home/nics/CSW/openpi/umi/data.zip eva14:/share/guqiuyi-local/umi_data_raw
6
7  # 服务器：将mp4和json处理成lerobot格式
8  unzip data.zip
9  python convert_mp4_data_to_lerobot_downsample.py --stage1_dir=/share/guqiuyi-local/umi_ready_to_upload/batch_pickup_cubes_0125 --repo=local/{task_name}

```



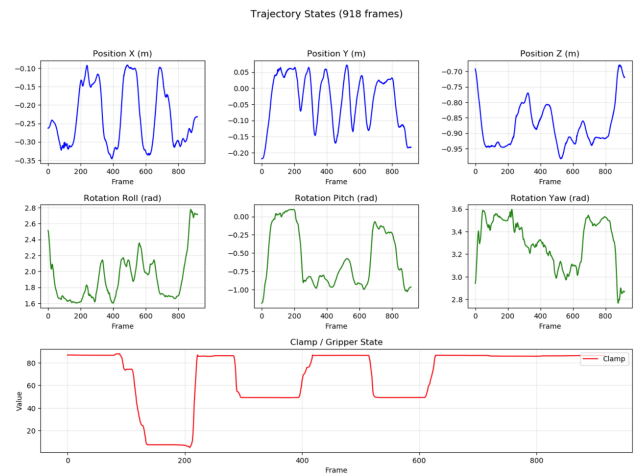
3D鼠标遥操数据

```

v fastumi_data
> batch_data_0125
v batch_data_0125_vis
  episode00000_states.png
  {} episode00000.json
  episode00000.mp4
  episode00000.serial.txt
  stage1_00000_try_250801DR48FP25002268.log

```

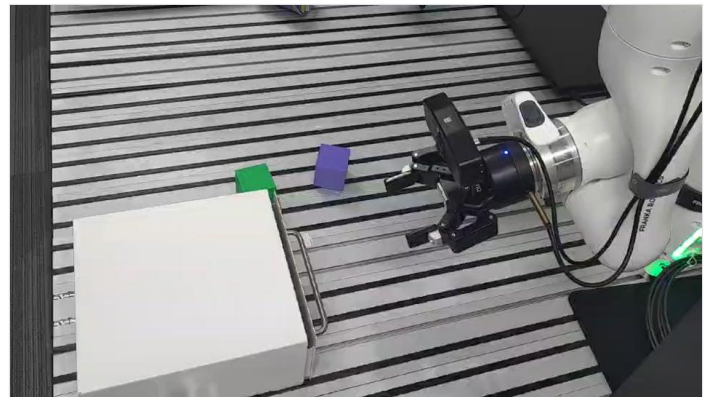
上传服务器的数据内容



umi数据

采放置方块任务的数据+训练

- 左边：遥操20条
- 右边：无本体76条



1.27-1.29

无本体数据 --> 遥操lerobot格式

process_umi_data_server.py

```

1  """
2  [Server Machine] High-Performance LeRobot Dataset Converter for UMI Data.
3  Features:
4  1. Schema matches Franka Teleop (tcp_pose, gripper_pose, fish_eye_front).
5  2. Action: 7D [pos(3), rotvec(3), absolute_gripper_width(1)].
6  3. Extremely fast video reading using imageio iterator.
7
8  Usage:
9      python process_umi_data_fast.py --stage1_dir=./umi_data --
      repo=local/umi_demo
  
```

```

10  """
11  import argparse
12  import json
13  import shutil
14  from pathlib import Path
15
16  import numpy as np
17  import imageio.v3 as iio
18  from scipy.spatial.transform import Rotation as R
19  from unittest.mock import patch
20
21  from lerobot.common.datasets.lerobot_dataset import HF_LEROBOT_HOME,
  LeRobotDataset
22
23  def pose7_to_pos_rotvec(pose7: np.ndarray) -> tuple[np.ndarray, np.ndarray]:
24      """pose7: [x,y,z,qx,qy,qz,qw] -> (pos(3,), rotvec(3,))"""
25      pos = pose7[:3].astype(np.float32)
26      quat = pose7[3:7].astype(np.float32) # (qx,qy,qz,qw)
27      rotvec = R.from_quat(quat).as_rotvec().astype(np.float32)
28      return pos, rotvec
29
30  def load_episode_json(json_path: Path):
31      meta = json.loads(json_path.read_text(encoding="utf-8"))
32      records = meta["records"]
33      if len(records) < 2:
34          raise ValueError(f"{json_path} has <2 records, cannot build episode.")
35      poses = np.asarray([r["pose"] for r in records], dtype=np.float32) # (T,7)
36
37      if "clamp" in records[0]:
38          clamp = np.asarray([r["clamp"] for r in records],
  dtype=np.float32).reshape(-1, 1) # (T,1)
39      else:
40          clamp = None
41
42      fps = float(meta.get("fps", 60.0))
43      return poses, clamp, fps, len(records)
44
45  def compute_stride(orig_fps: float, target_fps: float) -> tuple[int, float]:
46      """Return (stride, achieved_fps)."""
47      if target_fps <= 0:
48          raise ValueError("target_fps must be > 0")
49      stride = int(round(orig_fps / target_fps))
50      stride = max(1, stride)
51      achieved_fps = orig_fps / target_fps # 修正: 之前这里除错了, 应该是
  orig/stride
52      achieved_fps = orig_fps / stride
53      return stride, achieved_fps

```

```

54
55 def main(
56     stage1_dir: str,
57     repo_name: str,
58     robot_type: str = "franka_research_3",
59     target_fps: float = 10.0,
60     task_text: str = "put the purple cup into the plate",
61 ):
62     stage1_dir = Path(stage1_dir)
63     assert stage1_dir.exists(), f"stage1_dir not found: {stage1_dir}"
64
65     mp4_files = sorted(stage1_dir.glob("episode*.mp4"))
66     if not mp4_files:
67         raise FileNotFoundError(f"No episode*.mp4 found in {stage1_dir}")
68
69     first_json = stage1_dir / (mp4_files[0].stem + ".json")
70     if not first_json.exists():
71         raise FileNotFoundError(f"Missing json for first mp4: {first_json}")
72
73     poses0, clamp0, fps0, _ = load_episode_json(first_json)
74
75     # Downsample config
76     stride, achieved_fps = compute_stride(fps0, float(target_fps))
77     fps_int = int(round(achieved_fps))
78     fps_int = max(1, fps_int)
79
80     print(f"[INFO] Downsample: {fps0}Hz -> {achieved_fps:.4f}Hz (stride=
{stride}), dataset fps={fps_int}")
81
82     # Peek first frame for H,W
83     first_frame = io.imread(mp4_files[0], index=0)
84     if first_frame.ndim != 3 or first_frame.shape[2] != 3:
85         raise ValueError(f"Unexpected frame shape: {first_frame.shape}")
86     H, W, _ = first_frame.shape
87
88     # Prepare output repo (clean once)
89     out_path = HF_LEROBOT_HOME / repo_name
90     if out_path.exists():
91         shutil.rmtree(out_path)
92
93     use_clamp = clamp0 is not None
94     cam_key = "fish_eye_front"
95
96     # =====
97     # [修改 1] 对齐 Franka Teleop 格式的 Features
98     # =====
99     features = {

```

```

100         f"observation.images.{cam_key}": {
101             "dtype": "image",
102             "shape": (H, W, 3),
103             "names": ["height", "width", "channel"],
104         },
105         "observation.state.tcp_pose": {
106             "dtype": "float32",
107             "shape": (7,),
108             "names": ["x", "y", "z", "qx", "qy", "qz", "qw"],
109         },
110         "observation.state.gripper_pose": {
111             "dtype": "float32",
112             "shape": (1,),
113             "names": ["gripper"],
114         },
115         "action": {
116             "dtype": "float32",
117             "shape": (7,),
118             "names": ["x", "y", "z", "rx", "ry", "rz", "gripper_width"],
119         },
120         "task_idx": {"dtype": "int64", "shape": (1,), "names": None},
121         "subtask_idx": {"dtype": "int64", "shape": (1,), "names": None},
122         "is_failed": {"dtype": "bool", "shape": (1,), "names": None},
123         "reward": {"dtype": "float32", "shape": (1,), "names": None},
124         "return": {"dtype": "float32", "shape": (1,), "names": None},
125         "done": {"dtype": "bool", "shape": (1,), "names": None},
126     }
127
128     # === [修改 2] Monkey Patch mkdir 以解决 FileExistsError ===
129     original_mkdir = Path.mkdir
130     def patched_mkdir(self, mode=0o777, parents=False, exist_ok=False):
131         return original_mkdir(self, mode, parents, exist_ok=True)
132
133     with patch('pathlib.Path.mkdir', side_effect=patched_mkdir, autospec=True):
134         dataset = LeRobotDataset.create(
135             repo_id=repo_name,
136             robot_type=robot_type,
137             fps=fps_int,
138             features=features,
139             image_writer_threads=8,
140             image_writer_processes=4,
141         )
142     # =====
143
144     total_eps = 0
145     total_frames = 0
146

```

```

147     for mp4_path in mp4_files:
148         json_path = stage1_dir / (mp4_path.stem + ".json")
149         if not json_path.exists():
150             print(f"[WARN] Missing json for {mp4_path.name}, skip")
151             continue
152
153         poses, clamp, fps_meta, T = load_episode_json(json_path)
154
155         if use_clamp and clamp is None:
156             raise ValueError(f"{json_path} has no clamp but dataset expects
clamp.")
157
158         #
=====
159         # [核心继承] 保留了你第一个脚本的高效指针遍历逻辑
160         #
=====
161         ds_indices = np.arange(0, T, stride, dtype=np.int64)
162         T_ds = int(ds_indices.shape[0])
163         ds_ptr = 0
164         written = 0
165
166         # 流式读取视频，不把整个视频加载进内存
167         for i, frame in enumerate(io.iter(mp4_path)):
168             if i >= T:
169                 break
170             if ds_ptr >= T_ds or i != ds_indices[ds_ptr]:
171                 continue
172
173             i_obs = int(ds_indices[ds_ptr])
174             i_next = int(ds_indices[ds_ptr + 1]) if ds_ptr + 1 < T_ds else
i_obs
175
176             frame = np.asarray(frame, dtype=np.uint8)
177
178             # -----
179             # [修改 3] 映射到标准 State (tcp_pose 7D, gripper_pose 1D)
180             # -----
181             # tcp_pose: 直接使用 poses[i_obs] (里面是 x,y,z, qx,qy,qz,qw)
182             tcp_pose = poses[i_obs].astype(np.float32)
183
184             if use_clamp:
185                 gripper_pose = np.array([float(clamp[i_obs].item())],
dtype=np.float32)
186             else:
187                 gripper_pose = np.array([0.0], dtype=np.float32)
188

```



```

189         # -----
190         # [修改 4] 映射到标准 Action (pos 3D + rotvec 3D + ABSOLUTE clamp 1D)
191         # -----
192         pos_n, rotvec_n = pose7_to_pos_rotvec(poses[i_next])
193
194         if use_clamp:
195             # 动作直接用下一帧的绝对宽度
196             grip_n = np.array([float(clamp[i_next].item())],
dtype=np.float32)
197         else:
198             grip_n = np.array([0.0], dtype=np.float32)
199
200         action7 = np.concatenate([pos_n, rotvec_n, grip_n],
axis=0).astype(np.float32)
201
202         # -----
203         # [修改 5] 构建符合 Franka Teleop 格式的 Frame Dict
204         # -----
205         dataset.add_frame(
206             {
207                 f"observation.images.{cam_key}": frame,
208                 "observation.state.tcp_pose": tcp_pose,
209                 "observation.state.gripper_pose": gripper_pose,
210                 "action": action7,
211                 "task_idx": np.array([0], dtype=np.int64),
212                 "subtask_idx": np.array([0], dtype=np.int64),
213                 "is_failed": np.array([False], dtype=bool),
214                 "reward": np.array([0.0], dtype=np.float32),
215                 "return": np.array([0.0], dtype=np.float32),
216                 "done": np.array([ds_ptr == T_ds - 1], dtype=bool),
217                 "task": task_text,
218             }
219         )
220
221         written += 1
222         ds_ptr += 1
223
224         if written != T_ds:
225             raise RuntimeError(f"[STRICT ERROR] {mp4_path.name}: written=
{written} != T_ds={T_ds}.")
226
227         dataset.save_episode()
228         total_eps += 1
229         total_frames += written
230         print(f"[OK] {mp4_path.name} -> episode_saved, frames_used={written}")
231
232         print(f"\nDONE -> repo saved at: {out_path}")

```

```

233     print(f"episodes={total_eps}, total_frames_written={total_frames},
dataset_fps={fps_int}")
234
235 if __name__ == "__main__":
236     p = argparse.ArgumentParser()
237     p.add_argument("--input_dir", required=True, help="Directory containing
episodeXXXXX.mp4/json")
238     p.add_argument("--repo_id", required=True, help="Output LeRobot repo name")
239     p.add_argument("--robot_type", default="franka_research_3")
240     p.add_argument("--target_fps", type=float, default=10.0, help="Downsample
target fps, e.g. 10")
241     p.add_argument("--task_text", type=str, default="put cubes into the
cabinet")
242     args = p.parse_args()
243
244     main(
245         stage1_dir=args.input_dir,
246         repo_name=args.repo_id,
247         robot_type=args.robot_type,
248         target_fps=args.target_fps,
249         task_text=args.task_text,
250     )

```

统一训练脚本，测试是否ok --> 不能用三维表征旋转

配置名	配置	效果	归一化	配置脚本
pi05_xv_finetune_test1 目的：看一下时间是否变短	xv_policy .py脚本 batch_size=16 action_horizon=10 num_training_steps=20_000	果然训练只需要11个小时了 	"actions": { "mean": [0.00044734959 374181926, 0.01311508379 8766136, -0.01006749924 2722988, 0.96442139148 71216,	代码块 <pre> 1 TrainConfig(2 3 name="pi05_xv_finetune_test1", 4 5 model=pi0_config.Pi0Config(6 7 pi05=True, 8 9 action_dim=32, 10 # pi05 is trained with 32-dim actions 11 12 action_horizon </pre>

			-0.00966465007 5137615, 0.00150925479 82931137, 0.00859971158 2064629, 0.95871543884 27734, -0.00024088613 281492144, 77.2639770507 8125	=10, 7), 8 data=LeRobotXV DataConfig(9 # repo_id="/share/chenshuaiwen - local/.cache/h f_home/fastumi /0112", 10 repo_id="local /umi_pick_plac e_cubes_0125_c sw",
pi05_frank a_rel_finet une_test1	action第7 维是 clamp width, 需要归一 化 部署时候 注意 gripper给 阈值	有点操作的意思，但机械 臂还是容易莫名往上走 鉴于任务比较难且数据只 有76条，后面换简单任务 验证	"actions": { "mean": [0.00036757384 07764584, 0.01084199547 7676392, -0.00792767573 1480122, 0.00640954403 20670605, -0.00190663093 2353437, 0.00850575789 809227, 77.2639770507	代码块 1 TrainConfig(2 name="pi05_fra nka_rel_finetu ne_test1", 3 # Pi0.5 配 置 4 model=pi0_conf ig.Pi0Config(p i05=True, action_dim=32, action_horizon =10), 5 6 data=LeRobotFr ankaRelDataCon fig(7 repo_id="local /umi_pick_plac e_cubes_0128",

结论：模型输出需要用6d旋转 + 归一化后的gripper width

清理服务器文件

- `/share/guqiuyi-local`
 - `Assets`：存归一化结果
 - `Checkpoints`：存模型
 - `franka_ready_to_upload`：存上传到服务器的mp4等遥操数据
 - `umi_ready_to_upload`：存上传到服务器的mp4等无本体数据
 - `huggingface/lerobot/local`：存lerobot格式的数据

```
(openpi) (base) guqiuyi@eval4-guqiuyi:/$ df -h
Filesystem      Size  Used Avail Use% Mounted on
root-guqiuyi/containers/guqiuyi  99G   96G   3.2G  97% /
none            492K   4.0K  488K   1% /dev
tmpfs           100K    0   100K   0% /dev/txd
/dev/sda        21T   12T   9.3T  57% /share/public-local
10.1.0.2:/san_disk/users/guqiuyi  31T   29T   2.3T  93% /share/guqiuyi-nfs
10.1.0.2:/san_disk/users/public   31T   29T   2.3T  93% /share/public-nfs
/dev/loop27     30G   3.6M   30G   1% /var/lib/docker
/dev/sdb2       878G  315G  519G  38% /dev/nvidia0
tmpfs           100K    0   100K   0% /dev/.lxd-mounts
tmpfs           504G   12K  504G   1% /proc/driver/nvidia
tmpfs           504G  196K  504G   1% /dev/shm
tmpfs           202G  208K  202G   1% /run
tmpfs            5.0M    0   5.0M   0% /run/lock
tmpfs           48G    0   48G   0% /run/user/1321
```

鱼眼图像取消resize

在exp的config中加上 `IMAGE_SIZE = (1280, 1280)`

代码块

```
1  "observation.images.fish_eye_front": {
2      "dtype": "image",
3      "shape": [
4          240,
5          420,
6          3
7      ],
8      "names": [
9          "height",
10         "width",
11         "channel"
```

代码块

```
1  "observation.images.fish_eye_front": {
2      "dtype": "image",
3      "shape": [
4          1280,
5          1280,
6          3
7      ],
8      "names": [
9          "height",
10         "width",
11         "channel"
```

```
12     ]
13     },
```

```
12     ]
13     },
```

1.30-1.31

测试rel action的旋转表征和夹爪表征对模型性能的影响

遥操数据	v1	原来的3维旋转表征	用3维 franka_rel_policy_origin compute_norm_stats_skip_action7 train_config pi05_franka_rel_finetune wrapper_rel	ckpt_franka_rel_pick_place_0122 140条训
	v2	在policy脚本中将action input改成6d，归一化时skip第10维的gripper	用6维franka_rel_policy compute_norm_stats_skip_action10	
	v3	改lerobot格式，后面所有脚本都用xv的	用6维 xv_policy	
无本体数据	v4	原来的xv脚本	用6维 xv_policy	
	v5	改lerobot格式，变成		

采上货任务的数据+训练

- 无本体：300-条
- 遥操：



采pickplace杯子任务的数据+训练

- 无本体：200+224
- 遥操：

采微波炉加热食物任务的数据+训练

【takeaway】 1.15-1.31

1. 数据采集

- a. 怎么保证无本体数据的轨迹机械臂能完成？
- b. 3D鼠标操作难度
- c. 用UMI遥操franka（参考VR手柄遥操）

2. 数据处理

- a. 只有保证训练脚本和部署脚本统一，才能公平对比数据效率
- b. 6d比3d更容易学，后续全改成6d rot

3. 论文

- a. 短期能写个workshop paper，评估1) 无本体数据怎么用才好使，2) 能做哪些类型的task，3) 采集手法
- b. CoRL：
 - i. 数据角度：增加头部视角、双臂
 - ii. 算法角度：cotrain

2.1-2.9

更新SOP

明确分工 秋怡 顾秋怡

[数据](#) 数据采集-模型训练-任务评测SOP

拿放杯子任务进度

1. 无本体拿放杯子(400条 6个模型)

无本体数据（输入图像&rel rot6d，输出rel action）			无本体数据（输入图像，输出rel action）		
100条数据	200条数据	400条数据	100条数据	200条数据	400条数据
完成	完成	完成	完成	完成	完成
ckpt_umi_pick_place_cup_0130_eval100	ckpt_umi_pick_place_cup_0130_eval200	ckpt_umi_pick_place_cup_0130_eval400	ckpt_umi_pick_place_cup_0130_eval100_image	ckpt_umi_pick_place_cup_0130_eval200_image	ckpt_umi_pick_place_cup_0130_eval400_image
11/20 55%	14/20 70%	15/20 75%	1/20 5%	4.5/20 22.5%	18+/20 90+%

a. 训练 秋怡 顾秋怡

- ☒ ~~【2.2】~~ 训完5个不同setting
- ☒ ~~【2.3】~~ 少训的eval100_image训起来了
- ☐ 【2.8】 用224条数据训eval100_image、eval50_image、eval50、eval100

b. 部署测试 大为 赵大为

- ☒ ~~【2.2】~~ 测了4个模型
- ☒ ~~【2.3】~~ 测了eval200_image
- ☒ ~~【2.5】~~ 测1个模型eval100_image
- ☐ 【2.9】

2. 遥操拿放杯子（100条 4个模型）

遥操数据（输入图像&abs state，输出rel action rot6d）		遥操数据（输入图像，输出rel action rot6d）	
50条数据	100条数据	50条数据	100条数据
完成	完成	完成	完成
ckpt_franka_pick_place_cup_0202_eval50	ckpt_franka_pick_place_cup_0202_eval100	ckpt_franka_pick_place_cup_0202_eval50_image	ckpt_franka_pick_place_cup_0202_eval100_image
20/20 100%	20/20 100%	20/20 100%	20/20 100%

a. 数据采集 小龙 单小龙

- ☒ ~~【2.2】~~ 增加了20条，目前共有100条
- i. 完成了

b. 训练 秋怡 顾秋怡

☑️ ~~【2.2】~~ 训eval100模型

☑️ ~~【2.3】~~ 训eval100_image模型

代码块

```
1  export HF_LEROBOT_HOME=/share/guqiuyi-local/huggingface/lerobot
2  export HF_DATASETS_OFFLINE=1
3
4      TrainConfig(
5          name="pi05_franka_rel6d_image_only_finetune",
6
7          model=pi0_config.Pi0Config(
8              pi05=True,
9              action_dim=32,
10             action_horizon=10,
11             discrete_state_input=False # 无state输入!!
12         ),
13
14         data=LeRobotFrankaRel6dDataConfig(
15             # for pick-place-cup task
16             repo_id="local/franka_pick_place_cup_0202_eval100",
17
18             # for stock-shelves task
19             # repo_id="local/franka_stock_shelves_0201_eval50",
20             # repo_id="local/franka_stock_shelves_0202_eval100",
21             base_config=DataConfig(prompt_from_task=True),
22         ),
23
24         weight_loader=weight_loaders.CheckpointWeightLoader(
25             "/home/guqiuyi/.cache/openpi/openpi-
assets/checkpoints/pi05_base/params"
26         ),
27
28         num_train_steps=20_000,
29         batch_size=16,
30         save_interval=2000,
31         checkpoint_base_dir="/share/guqiuyi-local/checkpoints",
32         assets_base_dir="/share/guqiuyi-local/assets",
33     ),
34
35     CUDA_VISIBLE_DEVICES=3 uv run
scripts/compute_norm_stats_skip_action10.py --config-name
pi05_franka_rel6d_image_only_finetune
36
37     CUDA_VISIBLE_DEVICES=3 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
scripts/train.py pi05_franka_rel6d_image_only_finetune --exp-
```

```
name=ckpt_franka_pick_place_cup_0202_eval100_image --overwrite
```

☒ ~~【2.4】训eval50、eval100_image两模型~~

代码块

```
1  # ckpt_franka_pick_place_cup_0202_eval50
2      TrainConfig(
3          name="pi05_franka_rel6d_finetune",
4
5          model=pi0_config.Pi0Config(
6              pi05=True,
7              action_dim=32,
8              action_horizon=10
9          ),
10
11         data=LeRobotFrankaRel6dDataConfig(
12             # for pick-place-cup task
13             repo_id="local/franka_pick_place_cup_0202_eval50",
14             # repo_id="local/franka_pick_place_cup_0202_eval100",
15
16             # for stock-shelves task
17             # repo_id="local/franka_stock_shelves_0201_eval50",
18             # repo_id="local/franka_stock_shelves_0202_eval100",
19
20             # for open-laptop task
21             # repo_id="local/franka_open_laptop_0204_eval50",
22             # repo_id="local/franka_open_laptop_0204_eval100",
23             base_config=DataConfig(prompt_from_task=True),
24         ),
25
26         weight_loader=weight_loaders.CheckpointWeightLoader(
27             "/home/guqiuyi/.cache/openpi/openpi-
assets/checkpoints/pi05_base/params"
28         ),
29
30         num_train_steps=20_000,
31         batch_size=16,
32         save_interval=2000,
33         checkpoint_base_dir="/share/guqiuyi-local/checkpoints",
34         assets_base_dir="/share/guqiuyi-local/assets",
35     ),
36     CUDA_VISIBLE_DEVICES=2 uv run
scripts/compute_norm_stats_skip_action10.py --config-name
pi05_franka_rel6d_finetune
37
```

```

38  CUDA_VISIBLE_DEVICES=2 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
    scripts/train.py pi05_franka_rel6d_finetune --exp-
    name=ckpt_franka_pick_place_cup_0202_eval50 --overwrite
39
40  # ckpt_franka_pick_place_cup_0202_eval50_image
41      TrainConfig(
42          name="pi05_franka_rel6d_image_only_finetune",
43
44          model=pi0_config.Pi0Config(
45              pi05=True,
46              action_dim=32,
47              action_horizon=10,
48              discrete_state_input=False # 无state输入!!
49          ),
50
51          data=LeRobotFrankaRel6dDataConfig(
52              # for pick-place-cup task
53              repo_id="local/franka_pick_place_cup_0202_eval50",
54              # repo_id="local/franka_pick_place_cup_0202_eval100",
55
56              # for stock-shelves task
57              # repo_id="local/franka_stock_shelves_0201_eval50",
58              # repo_id="local/franka_stock_shelves_0202_eval100",
59
60              # for open-laptop task
61              # repo_id="local/franka_open_laptop_0204_eval50",
62              # repo_id="local/franka_open_laptop_0204_eval100",
63              base_config=DataConfig(prompt_from_task=True),
64          ),
65
66          weight_loader=weight_loaders.CheckpointWeightLoader(
67              "/home/guqiuyi/.cache/openpi/openpi-
assets/checkpoints/pi05_base/params"
68          ),
69
70          num_train_steps=20_000,
71          batch_size=16,
72          save_interval=2000,
73          checkpoint_base_dir="/share/guqiuyi-local/checkpoints",
74          assets_base_dir="/share/guqiuyi-local/assets",
75      ),
76  CUDA_VISIBLE_DEVICES=3 uv run
    scripts/compute_norm_stats_skip_action10.py --config-name
    pi05_franka_rel6d_image_only_finetune
77
78  CUDA_VISIBLE_DEVICES=3 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
    scripts/train.py pi05_franka_rel6d_image_only_finetune --exp-

```

```
name=ckpt_franka_pick_place_cup_0202_eval50_image --overwrite
```

i. 完成了

c. 部署测试  赵大为

☒ ~~[2.4]~~ 测eval100

上午eval100

```
1 ~/realRL$ bash run_server.sh
2
3 # 改~/qiuyi/openpi中1207行pi05_franka_rel6d_finetune的repo_id
4 repo_id="local/franka_pick_place_cup_0202_eval100"
5 ~/qiuyi/openpi$ uv run scripts/serve_policy.py policy:checkpoint --
  policy.config=pi05_franka_rel6d_finetune --
  policy.dir=/home/ubuntu/qiuyi/ckpts/ckpt_franka_pick_place_cup_0202_ev
  al100/19999
6
7 ~/realRL$ CYCLIC=false ./scripts/eval_policy.sh pick-place-cup 1
```

☒ ~~[2.6]~~ 测eval100_image和eval50_image

eval100_image

```
1 ~/realRL$ bash run_server.sh
2
3 # 改~/qiuyi/openpi中1207行pi05_franka_rel6d_image_only_finetune的
  repo_id
4 repo_id="local/franka_pick_place_cup_0202_eval100"
5 ~/qiuyi/openpi$ uv run scripts/serve_policy.py policy:checkpoint --
  policy.config=pi05_franka_rel6d_image_only_finetune --
  policy.dir=/home/ubuntu/qiuyi/ckpts/ckpt_franka_pick_place_cup_0202_ev
  al100_image/19999
6
7 ~/realRL$ CYCLIC=false ./scripts/eval_policy.sh pick-place-cup 20
```

eval50_image

```
1 100改成50
```

i. 完成了

上货任务进度

1. 无本体上货（400条 3个模型）

无本体数据（输入图像&rel rot6d, 输出rel action）			无本体数据（输入图像, 输出rel action） 注意这里action的z轴部分*1.5		
100条数据	200条数据	400条数据	200条数据	300条数据	400条数据
不训练	不训练	不训练	完成	完成	完成
			ckpt_umi_stock_shelves_0202_eval200_image	ckpt_umi_stock_shelves_0202_eval300_image	ckpt_umi_stock_shelves_0202_eval400_image
			12.5/20 62.5%	12.5/20 62.5%	11/20 55%

a. 数据采集 小龙 单小龙

- ☒ ~~【2.2】~~ 采了260+条
- ☒ ~~【2.3 2.4】~~ 采满400条

i. 完成了

b. 训练 秋怡 顾秋怡

- ☒ ~~【2.2】~~ 200条数据无state
- ☒ ~~【2.4】~~ 训练eval300_image和eval400_image

代码块

```
1  # ckpt_umi_stock_shelves_0202_eval300_image
2  TrainConfig(
3      name="pi05_umi_rel6d_image_only_finetune",
4      model=pi0_config.Pi0Config(
5          pi05=True,
6          action_dim=32, # pi05 is trained with 32-dim actions
7          action_horizon=10,
8          discrete_state_input=False # 无state输入!!
9      ),
10     data=LeRobotXVDataConfig(
11         # for pick-place-cup task
12         # repo_id="local/umi_pick_place_cup_0130_eval100",
13         # repo_id="local/umi_pick_place_cup_0130_eval200",
14         # repo_id="local/umi_pick_place_cup_0130_eval400",
15
16         # for stock-shelves task
17         # repo_id="local/umi_stock_shelves_0202_eval200",
18         repo_id="local/umi_stock_shelves_0202_eval300",
19         # repo_id="local/umi_stock_shelves_0202_eval400",
20         base_config=DataConfig(
21             prompt_from_task=True, # 用 dataset 的 "task" 字段做
22             prompt
23         ),
24     ),
25 )
```



```

24         weight_loader=weight_loaders.CheckpointWeightLoader(
25             "/home/guqiuyi/.cache/openpi/openpi-
assets/checkpoints/pi05_base/params"
26         ),
27
28         num_train_steps=20_000,
29         batch_size=16,
30         save_interval=2000,
31         checkpoint_base_dir="/share/guqiuyi-local/checkpoints",
32         assets_base_dir="/share/guqiuyi-local/assets",
33     ),
34
35     CUDA_VISIBLE_DEVICES=0 uv run scripts/compute_norm_stats.py --config-
name pi05_umi_rel6d_image_only_finetune
36
37     CUDA_VISIBLE_DEVICES=0 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
scripts/train.py pi05_umi_rel6d_image_only_finetune --exp-
name=ckpt_umi_stock_shelves_0202_eval300_image --overwrite
38
39
40     # ckpt_umi_stock_shelves_0202_eval300_image
41     TrainConfig(
42         name="pi05_umi_rel6d_image_only_finetune",
43         model=pi0_config.Pi0Config(
44             pi05=True,
45             action_dim=32, # pi05 is trained with 32-dim actions
46             action_horizon=10,
47             discrete_state_input=False # 无state输入!!
48         ),
49         data=LeRobotXVDataConfig(
50             # for pick-place-cup task
51             # repo_id="local/umi_pick_place_cup_0130_eval100",
52             # repo_id="local/umi_pick_place_cup_0130_eval200",
53             # repo_id="local/umi_pick_place_cup_0130_eval400",
54
55             # for stock-shelves task
56             # repo_id="local/umi_stock_shelves_0202_eval200",
57             # repo_id="local/umi_stock_shelves_0202_eval300",
58             repo_id="local/umi_stock_shelves_0202_eval400",
59             base_config=DataConfig(
60                 prompt_from_task=True, # 用 dataset 的 "task" 字段做
prompt
61             ),
62         ),
63         weight_loader=weight_loaders.CheckpointWeightLoader(
64             "/home/guqiuyi/.cache/openpi/openpi-
assets/checkpoints/pi05_base/params"

```

```

65         ),
66
67         num_train_steps=20_000,
68         batch_size=16,
69         save_interval=2000,
70         checkpoint_base_dir="/share/guqiuyi-local/checkpoints",
71         assets_base_dir="/share/guqiuyi-local/assets",
72     ),
73
74     CUDA_VISIBLE_DEVICES=1 uv run scripts/compute_norm_stats.py --config-
name pi05_umi_rel6d_image_only_finetune
75
76     CUDA_VISIBLE_DEVICES=1 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
scripts/train.py pi05_umi_rel6d_image_only_finetune --exp-
name=ckpt_umi_stock_shelves_0202_eval400_image --overwrite

```

i. 完成了

c. 部署测试  赵大为

i. 【2.3】试了下eval_200_image, pick不太行, place不错

☒ ~~【2.5】测3个模型eval200_image、eval300_image和eval400_image~~

i. 完成了

2. 遥操上货 (100条 4个模型)

遥操数据 (输入图像&abs state, 输出rel action rot6d)		遥操数据 (输入图像, 输出rel action rot6d)	
50条数据	100条数据	50条数据	100条数据
完成	完成	完成	完成
ckpt_franka_stock_shelves_0202_eval50	ckpt_franka_stock_shelves_0202_eval100	ckpt_franka_stock_shelves_0202_eval50_image	ckpt_franka_stock_shelves_0202_eval100_image
18/20 90%	18.5/20 92.5+%	13.5/20 67.5%	16/20 80%

a. 数据采集  单小龙

☒ ~~【2.2】增加了50条, 目前共有100条~~

i. 完成了

b. 训练  顾秋怡

☒ ~~【2.2】50条数据(del relrot3d relrot6d)都有state, 100条数据 (relrot6d)都有state~~

☒ ~~【2.3】训练eval50_image、eval100_image~~

代码块

```

1 export HF_LEROBOT_HOME=/share/guqiuyi-local/huggingface/lerobot
2 export HF_DATASETS_OFFLINE=1
3 # ckpt_franka_stock_shelves_0202_eval100_image

```

```

4     TrainConfig(
5         name="pi05_franka_rel6d_image_only_finetune",
6
7         model=pi0_config.Pi0Config(
8             pi05=True,
9             action_dim=32,
10            action_horizon=10,
11            discrete_state_input=False # 无state输入!!
12        ),
13
14        data=LeRobotFrankaRel6dDataConfig(
15            # for pick-place-cup task
16            # repo_id="local/franka_pick_place_cup_0202_eval100",
17
18            # for stock-shelves task
19            # repo_id="local/franka_stock_shelves_0201_eval50",
20            repo_id="local/franka_stock_shelves_0202_eval100",
21            base_config=DataConfig(prompt_from_task=True),
22        ),
23
24        weight_loader=weight_loaders.CheckpointWeightLoader(
25            "/home/guqiuyi/.cache/openpi/openpi-
assets/checkpoints/pi05_base/params"
26        ),
27
28        num_train_steps=20_000,
29        batch_size=16,
30        save_interval=2000,
31        checkpoint_base_dir="/share/guqiuyi-local/checkpoints",
32        assets_base_dir="/share/guqiuyi-local/assets",
33    ),
34
35    CUDA_VISIBLE_DEVICES=0 uv run
scripts/compute_norm_stats_skip_action10.py --config-name
pi05_franka_rel6d_image_only_finetune
36
37    CUDA_VISIBLE_DEVICES=0 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
scripts/train.py pi05_franka_rel6d_image_only_finetune --exp-
name=ckpt_franka_stock_shelves_0202_eval100_image --overwrite
38
39
40    # ckpt_franka_stock_shelves_0202_eval50_image
41    TrainConfig(
42        name="pi05_franka_rel6d_image_only_finetune",
43
44        model=pi0_config.Pi0Config(
45            pi05=True,

```

```

46         action_dim=32,
47         action_horizon=10,
48         discrete_state_input=False # 无state输入!!
49     ),
50
51     data=LeRobotFrankaRel6dDataConfig(
52         # for pick-place-cup task
53         # repo_id="local/franka_pick_place_cup_0202_eval100",
54
55         # for stock-shelves task
56         repo_id="local/franka_stock_shelves_0201_eval50",
57         # repo_id="local/franka_stock_shelves_0202_eval100",
58         base_config=DataConfig(prompt_from_task=True),
59     ),
60
61     weight_loader=weight_loaders.CheckpointWeightLoader(
62         "/home/guqiuyi/.cache/openpi/openpi-
assets/checkpoints/pi05_base/params"
63     ),
64
65     num_train_steps=20_000,
66     batch_size=16,
67     save_interval=2000,
68     checkpoint_base_dir="/share/guqiuyi-local/checkpoints",
69     assets_base_dir="/share/guqiuyi-local/assets",
70 ),
71 CUDA_VISIBLE_DEVICES=1 uv run
scripts/compute_norm_stats_skip_action10.py --config-name
pi05_franka_rel6d_image_only_finetune
72
73 CUDA_VISIBLE_DEVICES=1 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
scripts/train.py pi05_franka_rel6d_image_only_finetune --exp-
name=ckpt_franka_stock_shelves_0202_eval50_image --overwrite

```

i. 完成了

c. 部署测试  赵大为

☒ ~~【2.3】测了下eval100，效果不错~~

☒ ~~【2.4】测eval50、eval100~~

上午eval100

```

1  bash run_server.sh
2
3  # 改~/qiuyi/openpi中1207行pi05_franka_rel6d_finetune的repo_id
4  repo_id="local/franka_stock_shelves_0201_eval100"

```

```
5
6 uv run scripts/serve_policy.py policy:checkpoint --
  policy.config=pi05_franka_rel6d_finetune --
  policy.dir=/home/ubuntu/qiuyi/ckpts/ckpt_franka_stock_shelves_0202_eva
  l100/19999
7
8 CYCLIC=false ./scripts/eval_policy.sh stock-shelves 1
```

上午eval50

1 100区域改成50

☒ ~~【2.5】测eval50_image、eval100_image~~

i. 完成了

开笔记本电脑任务进度

1. 无本体开笔记本（400条 3个模型）

无本体数据（输入图像&rel rot6d，输出rel action）			无本体数据（输入图像，输出rel action）		
100条数据	200条数据	400条数据	200条数据	300条数据	400条数据
不训练	不训练	不训练			

a. 数据采集  单小龙

☒ ~~【2.5 2.6】采400条~~

☐ 完成了

b. 训练  顾秋怡

☐ 【2.8】训eval200_image、eval300_image、eval400_image

代码块

```
1 # ckpt_umi_open_laptop_0208_eval200_image
2 CUDA_VISIBLE_DEVICES=0 uv run scripts/compute_norm_stats.py --config-
  name pi05_umi_rel6d_image_only_finetune
3 CUDA_VISIBLE_DEVICES=0 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
  scripts/train.py pi05_umi_rel6d_image_only_finetune --exp-
  name=ckpt_umi_open_laptop_0208_eval200_image --overwrite
4
```

```

5 # ckpt_umi_open_laptop_0208_eval300_image
6 CUDA_VISIBLE_DEVICES=1 uv run scripts/compute_norm_stats.py --config-
  name pi05_umi_rel6d_image_only_finetune
7 CUDA_VISIBLE_DEVICES=1 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
  scripts/train.py pi05_umi_rel6d_image_only_finetune --exp-
  name=ckpt_umi_open_laptop_0208_eval300_image --overwrite
8
9 # ckpt_umi_open_laptop_0208_eval400_image
10 CUDA_VISIBLE_DEVICES=2 uv run scripts/compute_norm_stats.py --config-
  name pi05_umi_rel6d_image_only_finetune
11 CUDA_VISIBLE_DEVICES=2 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
  scripts/train.py pi05_umi_rel6d_image_only_finetune --exp-
  name=ckpt_umi_open_laptop_0208_eval400_image --overwrite

```

c. 部署测试 赵大为

☐ 【2.9】

2. 遥操开笔记本（100条 4个模型）

遥操数据（输入图像&abs state，输出rel action rot6d）		遥操数据（输入图像，输出rel action rot6d）	
50条数据	100条数据	50条数据	100条数据
完成	完成	完成	完成
ckpt_franka_open_laptop_0204_eval50	ckpt_franka_open_laptop_0204_eval100	ckpt_franka_open_laptop_0204_eval50_image	ckpt_franka_open_laptop_0204_eval100_image
13.5/20 67.5%	18.5/20 92.5%	18/20 90%	20/20 100%

a. 训练 顾秋怡

☒ ~~【2.4】训eval50、eval50_image、eval100、eval100_image~~

代码块

```

1 # ckpt_franka_open_laptop_0204_eval50
2 CUDA_VISIBLE_DEVICES=4 uv run
  scripts/compute_norm_stats_skip_action10.py --config-name
  pi05_franka_rel6d_finetune
3 CUDA_VISIBLE_DEVICES=4 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
  scripts/train.py pi05_franka_rel6d_finetune --exp-
  name=ckpt_franka_open_laptop_0204_eval50 --overwrite
4
5 # ckpt_franka_open_laptop_0204_eval50_image
6 CUDA_VISIBLE_DEVICES=6 uv run
  scripts/compute_norm_stats_skip_action10.py --config-name
  pi05_franka_rel6d_image_only_finetune
7 CUDA_VISIBLE_DEVICES=6 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
  scripts/train.py pi05_franka_rel6d_image_only_finetune --exp-

```

```

name=ckpt_franka_open_laptop_0204_eval50_image --overwrite
8
9 # ckpt_franka_open_laptop_0204_eval100
10 CUDA_VISIBLE_DEVICES=5 uv run
scripts/compute_norm_stats_skip_action10.py --config-name
pi05_franka_rel6d_finetune
11 CUDA_VISIBLE_DEVICES=5 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
scripts/train.py pi05_franka_rel6d_finetune --exp-
name=ckpt_franka_open_laptop_0204_eval100 --overwrite
12
13 # ckpt_franka_open_laptop_0204_eval100_image
14 CUDA_VISIBLE_DEVICES=7 uv run
scripts/compute_norm_stats_skip_action10.py --config-name
pi05_franka_rel6d_image_only_finetune
15 CUDA_VISIBLE_DEVICES=7 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
scripts/train.py pi05_franka_rel6d_image_only_finetune --exp-
name=ckpt_franka_open_laptop_0204_eval100_image --overwrite

```

b. 部署测试 赵大为

☒ ~~【2.6】~~ 测4个模型

eval_xx

```

1 ~/realRL$ bash run_server.sh
2
3 # 改~/qiuyi/openpi中1207行pi05_franka_rel6d_finetune的repo_id
4 repo_id="local/franka_open_laptop_0204_eval100/50"
5 ~/qiuyi/openpi$ uv run scripts/serve_policy.py policy:checkpoint --
policy.config=pi05_franka_rel6d_finetune --
policy.dir=/home/ubuntu/qiuyi/ckpts/ckpt_franka_open_laptop_0204_eval1
00/50/19999
6
7 ~/realRL$ CYCLIC=false ./scripts/eval_policy.sh pick-place-cup 20

```

eval_xx_image

```

1 ~/realRL$ bash run_server.sh
2
3 # 改~/qiuyi/openpi中1207行pi05_franka_rel6d_image_only_finetune的
repo_id
4 repo_id="local/franka_open_laptop_0204_eval100/50"
5 ~/qiuyi/openpi$ uv run scripts/serve_policy.py policy:checkpoint --
policy.config=pi05_franka_rel6d_image_only_finetune --
policy.dir=/home/ubuntu/qiuyi/ckpts/ckpt_franka_open_laptop_0204_eval1
00/50_image/19999

```


6

```
7 ~/realRL$ CYCLIC=false ./scripts/eval_policy.sh pick-place-cup 20
```

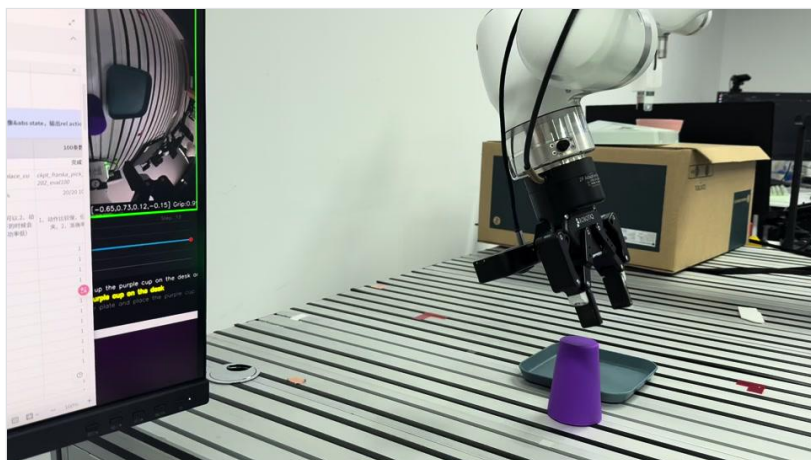
- 完成了

Demo

- 任务1: 拿放杯子

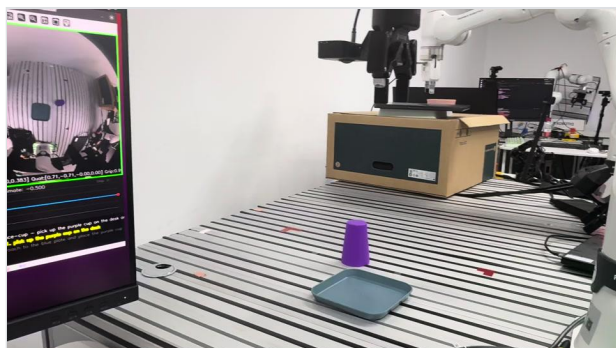
左遥操，右无本体

- 无本体
 - 训练输入的state和推理的state不一致，导致加入state后成功率低
 - 仅用图像观测这个任务是足够的
 - 频繁抓空跟设置的阈值有关，目前已经改善了一些



- 遥操

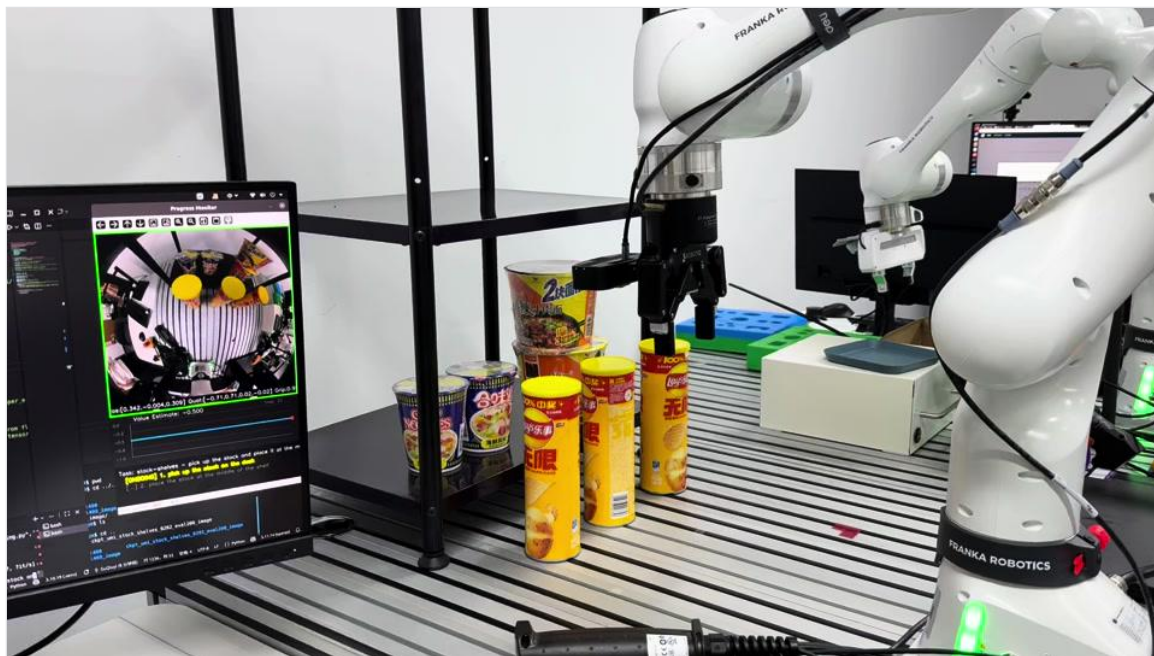
- 稳稳成功



- 任务2：上货

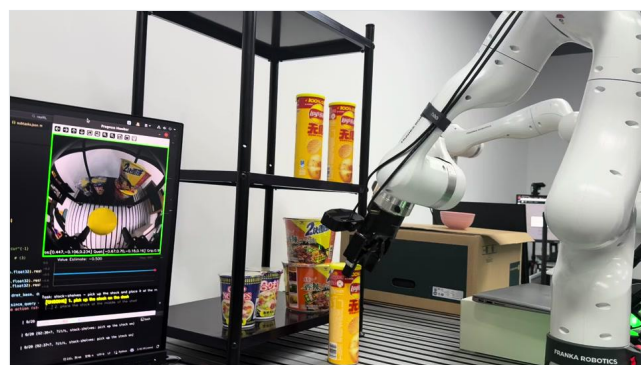
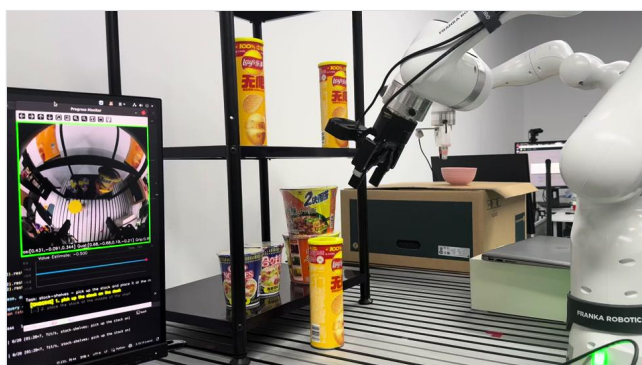
- 遥操

- eval100效果很好，并且能做连续上货（没这么采集数据）
 - eval100_image容易还没到达预期高度提前放下



- 无本体

- eval400_image同样存在还没到达预期高度提前放下的问题，目前通过 $\text{action_z} \times 1.5$ 解决
 - eval400_image修正夹爪开合的逻辑



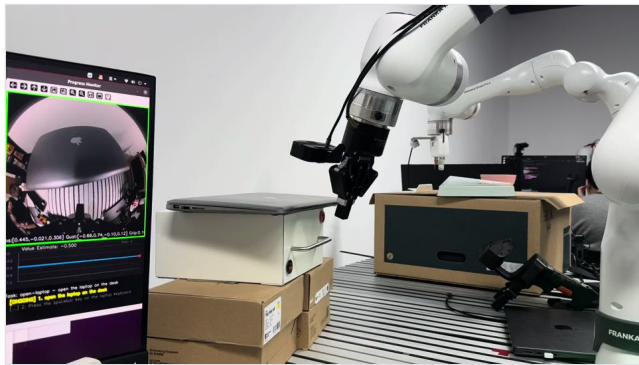
- 任务3：开笔记本

- 遥操

- 开笔记本一直很顺利，最后一步按空格键有时会偷懒



- 无本体
 - 不垫高会经常爆黄灯，需要在x轴方向action_x*1.5
 - eval400_image才能做这个任务



【takeaway】 2.1-2.12

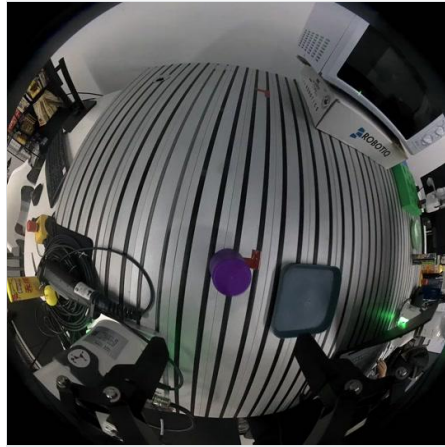
1. 无本体数据

- a. 目前来看简单任务只输入图像是能够完成的，和遥操数据相比数据效率是1:4?
- b. 拿放杯子和开笔记本在无本体100-300条时候的成功率太低，怀疑是部分数据有问题，增加实验：
 - i. Pick-place-cup
 - ☒ 分开来训模型
 - ii. Open-laptop
 - ☒ 400条拆分成200+200，训两个200的模型
 - iii. Stock-shelves
 - ☐ 加relative state，训一个400的模型
 - ☐ 加第三视角，重新采一波

2.10-2.12

不同人采的无本体数据质量评估

csw200----long224-----long100



- 增加实验：

- Pick-place-cup

- 只用龙哥的224条umi_pickup_cup_0130_224episode，训两个模型 瞎子

- `repo_id = "local/umi_pick_place_cup_0208_eval100/200"`

- `exp-name=ckpt_umi_pick_place_cup_0208_eval100_image`

- `exp-name=ckpt_umi_pick_place_cup_0208_eval200_image`

- 只用龙哥新采的100条，训一个模型 瞎子

- `repo_id = "local/umi_pick_place_cup_0209_eval100"` （降采样到10Hz）

- `exp-name=ckpt_umi_pick_place_cup_0209_eval100_image`

- 只用帅文的200条，训一个模型 成功率高

- `repo_id = "local/umi_pick_place_cup_0121"` （降采样到10Hz）

- `exp-name=ckpt_umi_pick_place_cup_0209_eval200_image`

- 只用帅文的100条，训一个模型 成功率也可以

- `repo_id = "local/umi_pick_place_cup_0210_eval100"` （降采样到10Hz）

- `exp-name=ckpt_umi_pick_place_cup_0210_eval100_image`

- Open-laptop

- 400episodes合在一起 勉强能打开

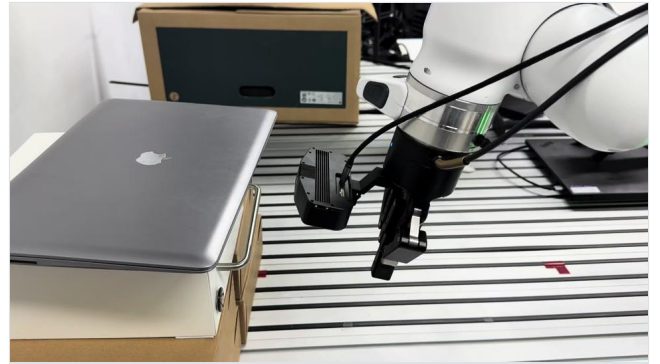
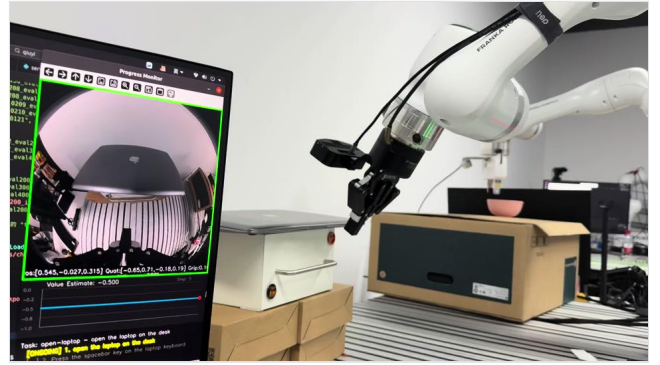
- `repo_id`

- `local/umi_open_laptop_0208_`

- `eval400_image`

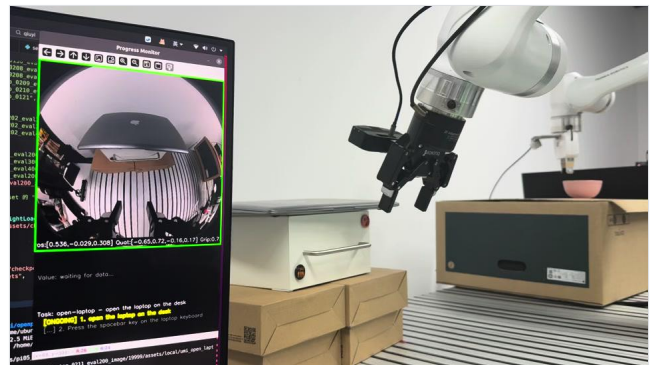
- 数据集umi_open-laptop_0204_100episodes + umi_open-laptop_0205_100episodes 有开笔记本的趋势，差一点点就接触到了

- repo_id
local/umi_open_laptop_0210_eval200_image
- exp-
name=ckpt_umi_open_laptop_0210_eval200_image



- 数据集umi_open-laptop_0206_200episodes 在玩抽象，有时还会表演缩骨功

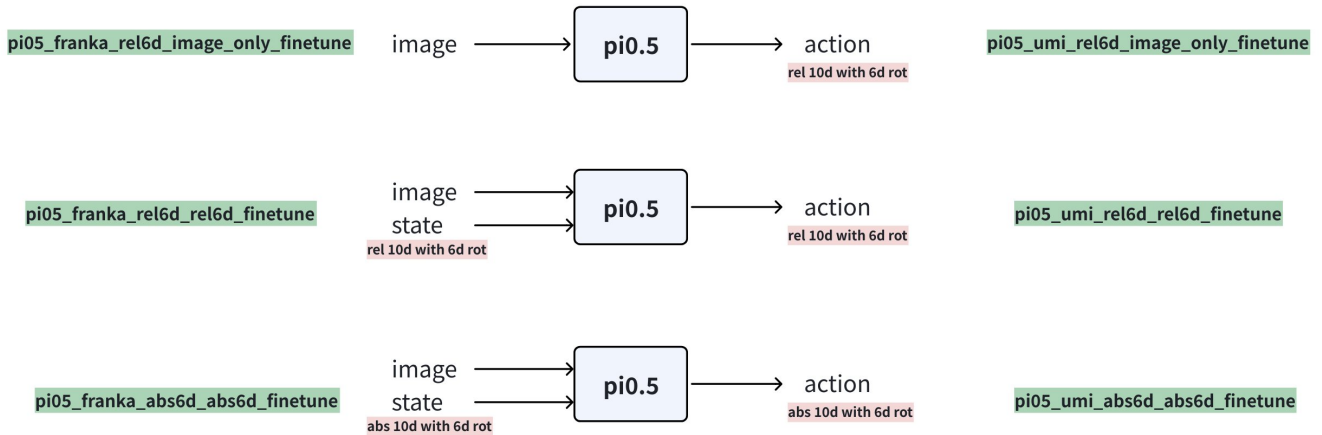
- repo_id
local/umi_open_laptop_0211_eval200_image
- exp-
name=ckpt_umi_open_laptop_0211_eval200_image



统一训练配置和脚本

Fine-tuning Franka Teleop configs.
franka_hybrid_policy.py

Fine-tuning FastUMI configs.
umi_hybrid_policy.py



#838 申朝阳

- 先前还有如下配置
 - pi05_franka_del_finetune : image + state (abs 8d with 4d quant) + action (delta 7d with 3d rot)
 - pi05_franka_rel3d_finetune : image + state (abs 8d with 4d quant) + action (rel 7d with 3d rot)
 - pi05_franka_rel6d_finetune : image + state (abs 8d with 4d quant) + action (rel 10d with 6d rot)
 - pi05_xv_finetune : image + state (rel 6d rot) + action (rel 10d with 6d rot)

• lerobot格式

- 遥操
 - state用四元数表示, gripper是脉冲信号, action是base坐标系下的delta action用3d rot

☒ 增加"demo_start_tcp_pose"

遥操

```
1  "observation.images.{cam}" = {"dtype": "image", "shape": (h, w, c),
    "names": ["height", "width", "channel"]},
2
3  "observation.state.tcp_pose": {"dtype": "float32", "shape": (7,),
    "names": ["x", "y", "z", "qx", "qy", "qz", "qw"]},
4  "observation.state.gripper_pose": {"dtype": "float32", "shape": (1,),
    "names": ["gripper"]},
5  "observation.state.demo_start_tcp_pose": {"dtype": "float32",
    "shape": (7,), "names": ["x", "y", "z", "qx", "qy", "qz", "qw"]},
```

```

6
7  "action": {"dtype": "float32", "shape": (7,), "names": ["dx", "dy",
  "dz", "drx", "dry", "drz", "gripper"]},
8
9  "task_idx": {"dtype": "int64", "shape": (1,), "names": None},
10 "subtask_idx": {"dtype": "int64", "shape": (1,), "names": None},
11 "is_failed": {"dtype": "bool", "shape": (1,), "names": None},
12 "reward": {"dtype": "float32", "shape": (1,), "names": None},
13 "return": {"dtype": "float32", "shape": (1,), "names": None},
14 "done": {"dtype": "bool", "shape": (1,), "names": None},
15 "task":

```

- 无本体

- state用3d rot表示，gripper是连续的夹爪宽度（mm为单位），action是下一帧的state

无本体

```

1  "image": {"dtype": "image", "shape": (H, W, 3), "names": ["height",
  "width", "channel"]},
2
3  "eef_pos": {"dtype": "float32", "shape": (3,), "names": ["x", "y",
  "z"]},
4  "eef_rot_axis_angle": {"dtype": "float32", "shape": (3,), "names":
  ["rx", "ry", "rz"]},
5  "gripper_width": {"dtype": "float32", "shape": (1,), "names":
  ["gripper_width"]},
6  "demo_start_pose": {"dtype": "float32", "shape": (6,), "names": ["x",
  "y", "z", "rx", "ry", "rz"]},
7
8  "actions": {"dtype": "float32", "shape": (7,), "names": ["x", "y",
  "z", "rx", "ry", "rz", "gripper_width"]},
9
10 "episode_start_idx": {"dtype": "int64", "shape": (1,), "names":
  ["episode_start_idx"]},
11 "episode_end_idx": {"dtype": "int64", "shape": (1,), "names":
  ["episode_end_idx"]},
12 "task":

```

- 测试

- pi05_franka_abs6d_abs6d_finetune

- repo_id="local/franka_pick_place_cup_0202_eval100"
- exp-name=ckpt_franka_pick_place_cup_0212_abs6d_abs6d


```

1 代码块
   CUDA_VISIBLE_DEVICES=0 uv run
   scripts/compute_norm_stats_skip_action10.py --config-name
   pi05_franka_abs6d_abs6d_finetune
2
3  CUDA_VISIBLE_DEVICES=0 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
   scripts/train.py pi05_franka_abs6d_abs6d_finetune --exp-
   name=ckpt_franka_pick_place_cup_0212_abs6d_abs6d --overwrite

```

- pi05_franka_rel6d_rel6d_finetune

- repo_id="local/franka_pick_place_cup_0212_eval100" (加了 demo_start_tcp_pose)
- exp-name=ckpt_franka_pick_place_cup_0212_rel6d_rel6d

代码块

```

1  python convert_franka_data_to_lerobot.py \
2  --input_dirs=/share/guqiuyi-local/franka_ready_to_upload/pick-place-
   cup_0202_100episodes \
3  --repo_id=local/franka_pick_place_cup_0212_eval100 \
4  --num_episodes=100 \
5  --seed=42
6
7  CUDA_VISIBLE_DEVICES=1 uv run
   scripts/compute_norm_stats_skip_action10.py --config-name
   pi05_franka_rel6d_rel6d_finetune
8
9  CUDA_VISIBLE_DEVICES=1 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
   scripts/train.py pi05_franka_rel6d_rel6d_finetune --exp-
   name=ckpt_franka_pick_place_cup_0212_rel6d_rel6d --overwrite

```

- pi05_franka_rel6d_image_only_finetune

- repo_id="local/franka_pick_place_cup_0212_eval100" (加了 demo_start_tcp_pose)
- exp-name=ckpt_franka_pick_place_cup_0212_rel6d_image_only

代码块

```

1  export HF_LEROBOT_HOME=/share/guqiuyi-local/huggingface/lerobot
2  export HF_DATASETS_OFFLINE=1
3
4  CUDA_VISIBLE_DEVICES=2 uv run
   scripts/compute_norm_stats_skip_action10.py --config-name
   pi05_franka_rel6d_image_only_finetune

```

```

5
6  CUDA_VISIBLE_DEVICES=2 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
   scripts/train.py pi05_franka_rel6d_image_only_finetune --exp-
   name=ckpt_franka_pick_place_cup_0212_rel6d_image_only --overwrite

```

- pi05_umi_rel6d_rel6d_finetune

- repo_id = "local/umi_pick_place_cup_0121"
- exp-name=ckpt_umi_pick_place_cup_0212_rel6d_rel6d

代码块

```

1  export HF_LEROBOT_HOME=/share/guqiuyi-local/huggingface/lerobot
2  export HF_DATASETS_OFFLINE=1
3
4  CUDA_VISIBLE_DEVICES=3 uv run scripts/compute_norm_stats.py --config-
   name pi05_umi_rel6d_rel6d_finetune
5
6  CUDA_VISIBLE_DEVICES=3 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
   scripts/train.py pi05_umi_rel6d_rel6d_finetune --exp-
   name=ckpt_umi_pick_place_cup_0212_rel6d_rel6d --overwrite

```

- pi05_umi_rel6d_image_only_finetune

- repo_id = "local/umi_pick_place_cup_0121",
- exp-name=ckpt_umi_pick_place_cup_0212_rel6d_image_only

代码块

```

1  CUDA_VISIBLE_DEVICES=4 uv run scripts/compute_norm_stats.py --config-
   name pi05_umi_rel6d_image_only_finetune
2
3  CUDA_VISIBLE_DEVICES=4 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
   scripts/train.py pi05_umi_rel6d_image_only_finetune --exp-
   name=ckpt_umi_pick_place_cup_0212_rel6d_image_only --overwrite

```

统一推理部署，提realRL PR

openpi: websocket_policy_server.py

☒ realRL里的wrapper

- tcp pose type

- Delta action
 - Abs action
 - Rel action
 - gripper type
 - 归一化的width
 - 脉冲信号
- ☐ openpi里的serve_policy
- 拿到观测obs = msgpack_numpy.unpackb(await websocket.recv()), 需要转成模型的input key

统一数据集，上传无问服务器

[回 数据采集列表](#)

表-2.9

- 【科研计划1】 为什么帅文和龙哥采的数据训出来效果不一样，怎么采数据是最优的？
- 【已解决】 为什么xv_policy训出来效果比umi_hybrid_policy好
- 【科研计划1】 加了rel state效果是变好还是变坏

1. 测新训的4个pick_place_cup_eval100/200
- a. hybrid和xv pocliy一致了，但是龙哥的100条数据还是训不出来
- ☐ 龙哥新采200条，测pick_place_cup_eval400，得达到100%成功率
- ☐ check2.4和2.6采集的open_laptop数据的区别

Pick-place-cup	无本体（输入图像，输出rel action）			无本体（输入图像&abs state，输出rel action）		
	100条	200条	400条	100条	200条	400条
	15.5/20 77.5%	18/20 90%	<u>18/20 90%</u>			
	遥操（输入图像，输出rel action）			遥操（输入图像&abs state，输出rel action）		

	50条	100条		50条	100条	
	20/20 100%	20/20 100%		20/20 100%	20/20 100%	
Stock-shelves	无本体（输入图像，输出rel action）			无本体（输入图像&abs state，输出rel action）		
	200条	300条	400条	100条	200条	400条
	12.5/20 62.5%	12.5/20 62.5%	14/20 70%			
	遥操（输入图像，输出rel action）			遥操（输入图像&abs state，输出rel action）		
	50条	100条		50条	100条	
	13.5/20 67.5%	16/20 80%		18/20 90%	18.5/20 92.5+%	
Open-laptop	无本体（输入图像，输出rel action）			无本体（输入图像&abs state，输出rel action）		
	200条	300条	400条	100条	200条	400条
	<u>0/20 0%</u>	<u>0/20 0%</u>	<u>10/20 50%</u>			
	遥操（输入图像，输出rel action）			遥操（输入图像&abs state，输出rel action）		
	50条	100条		50条	100条	
	18/20 90%	20/20 100%		16.5/20 82.5%	18.5/20 92.5%	

Pick-place-cup	无本体（输入图像，输出rel action）			遥操（输入图像，输出rel action）		遥操（输入图像&abs state，输出rel action）	
	100条	200条	400条	50条	100条	50条	100条
	15.5/20 77.5%	18/20 90%	<u>18/20 90%</u>	20/20 100%	20/20 100%	20/20 100%	20/20 100%
Stock-shelves	无本体（输入图像，输出rel action）			遥操（输入图像，输出rel action）		遥操（输入图像&abs state，输出rel action）	

	200条	300条	400条	50条	100条	50条	100条
	12.5/20 62.5%	12.5/20 62.5%	14/20 70%	13.5/20 67.5%	16/20 80%	18/20 90%	18.5/20 92.5+%
Open-laptop	无本体（输入图像，输出rel action）			遥操（输入图像，输出rel action）		遥操（输入图像&abs state，输出rel action）	
	200条	300条	400条	50条	100条	50条	100条
	<u>0/20 0%</u>	<u>0/20 0%</u>	<u>10/20 50%</u>	18/20 90%	20/20 100%	16.5/20 82.5%	18.5/20 92.5%

2.24

为什么两个人采集数据会有较大差异，定位在数据问题？代码问题？训练问题？

A是csw的200条，B是龙哥的100条

长度均值：A 258，B 299

timestamp fps：A 60.26，B 60.47

Δpos p99 median：A 0.0131m，B 0.0092m

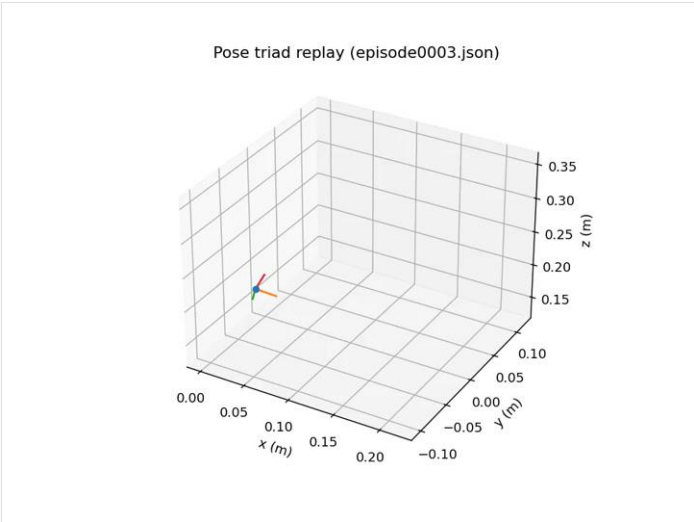
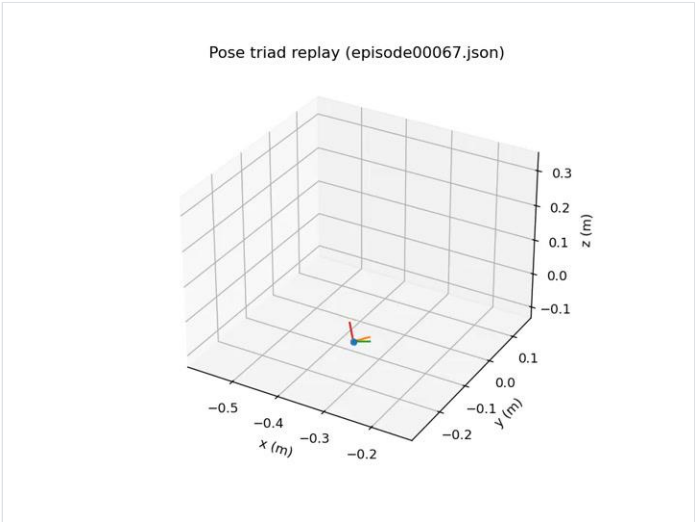
→ A 的平移步长约大 1.43 倍

Δrot p99 median：A 0.0269rad，B 0.0121rad

→ A 的旋转步长约大 2.2 倍

只是csw采的速度更快一点，旋转更多一点

坐标可视化出来也没问题，左csw，右龙哥



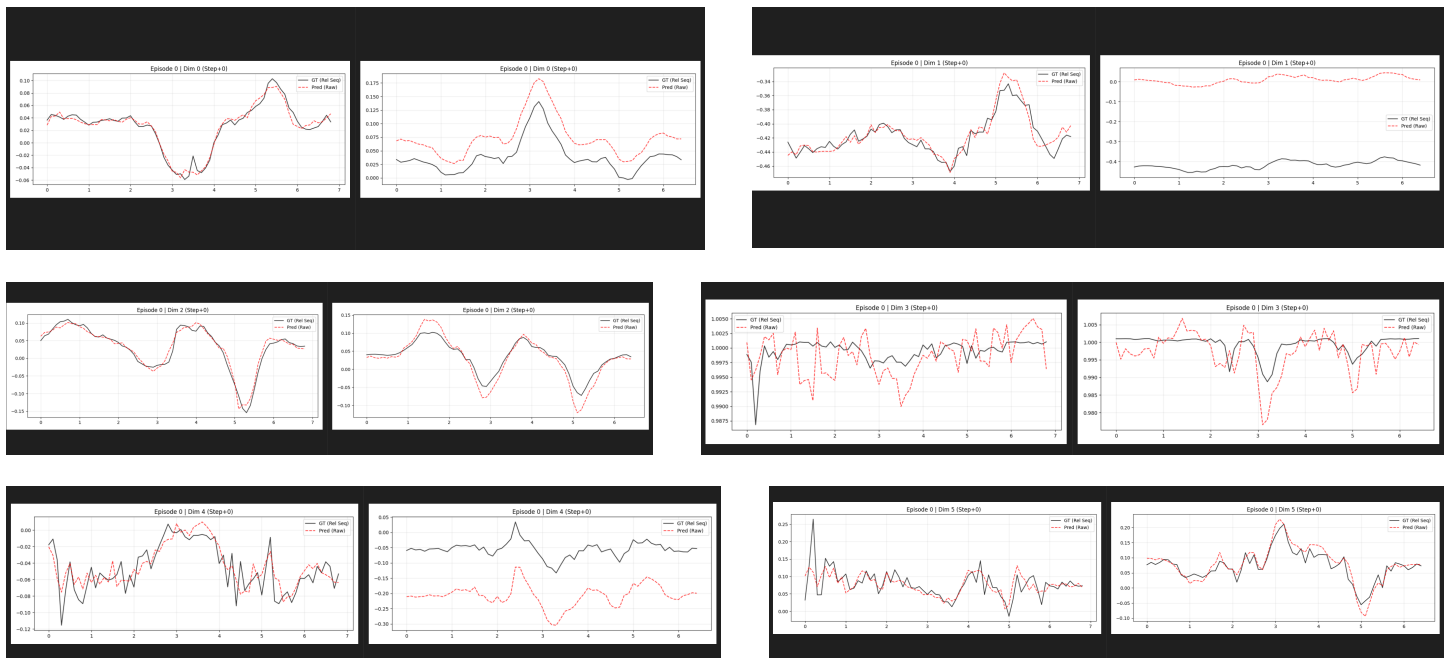
看是不是遇到了偶发bug或者某些参数没设置对

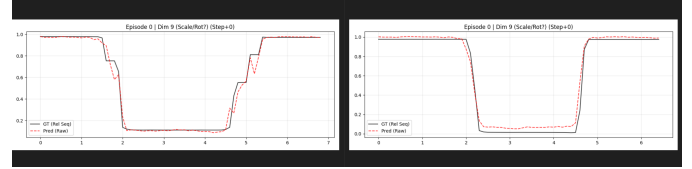
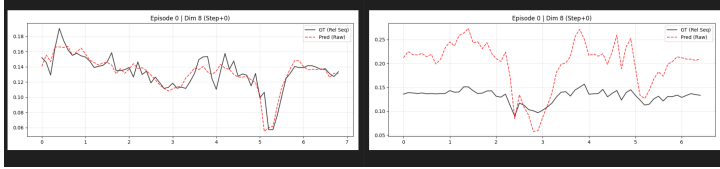
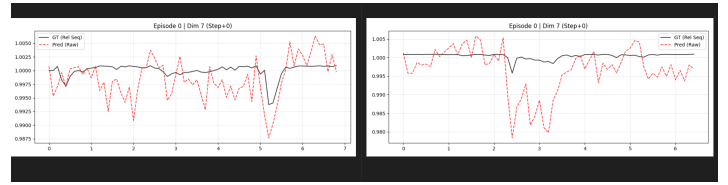
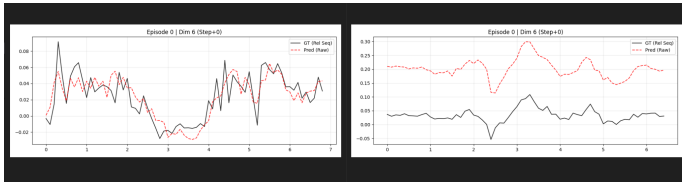
setting4: long数据+umi_hybrid_policy

鬼故事1.0

→ A 的旋转步长约为 2.2 倍

龙哥数据offline发现





这批数据做offline结果相似，dim00，01，04，06像是有一个固定的bias，08没学到
能用数据与这批数据的mean和std相差并不是很大

TODO：看看归一化啥情况，在哪做的，怎么做的

[openpi/transform.py](#)

有z-score归一化： $(x - \text{mean}) / (\text{std} + 1e-6)$

quantiles归一化： $(x - q01) / (q99 - q01 + 1e-6) * 2 - 1$ ，（现在使用，所以显式padding 0后会被拉到-1）

含义：把数据的 1% 分位映射到 -1，99% 分位映射到 +1（线性缩放）。这样对 heavy-tail/outlier 更鲁棒。

transform：

repack_transforms → data_transforms → Normalize → model_transforms

```

def create(self, assets_dirs: pathlib.Path, model_config: _model.BaseModelConfig) -> DataConfig:
    repack_transform = _transforms.Group(
        inputs=[
            _transforms.RepackTransform(
                {
                    # 图像映射: (3, 1280, 1280)
                    "image": "image",

                    # # 状态映射
                    "eef_pos": "eef_pos",
                    "eef_rot_axis_angle": "eef_rot_axis_angle",
                    "gripper_width": "gripper_width",
                    "demo_start_pose": "demo_start_pose",

                    # 动作映射
                    "actions": "actions",

                    # 提示词
                    "task": "task",
                }
            )
        ]
    )

    # We assume joint *velocity* actions, so we should *not* apply an additional delta transform.
    data_transforms = _transforms.Group(
        inputs=[xv_policy.XVInputs(model_type=model_config.model_type, action_dim=model_config.action_dim, action_type=model_config.action_type)],
        outputs=[xv_policy.XVOutputs()],
    )

    model_transforms = ModelTransformFactory()(model_config)

```

上边是能用的数据，下边是不能用的

代码块

```

1  {
2      "norm_stats": {
3          "state": {
4              "mean": [
5                  0.9721017479896545,
6                  0.06438173353672028,
7                  -0.0059456718154251575,
8                  -0.06447111070156097,
9                  0.894965648651123,
10                 0.23852209746837616
11             ],
12             "std": [
13                 0.03908156976103783,
14                 0.1880345195531845,
15                 0.118123859167099,
16                 0.17058105766773224,
17                 0.12695899605751038,
18                 0.30458253622055054
19             ],
20             "q01": [
21                 0.8189763835430145,

```

```
22         -0.4036883324146271,
23         -0.3675311414480209,
24         -0.4601298803329468,
25         0.4534862380940467,
26         -0.45881888012886046
27     ],
28     "q99": [
29         0.9999275063991546,
30         0.5329318737506865,
31         0.2585261615037918,
32         0.39639805269241335,
33         0.9998013868888841,
34         0.8570783271431923
35     ]
36 },
37     "actions": {
38         "mean": [
39             -0.003382311202585697,
40             0.041226886212825775,
41             -0.04003903269767761,
42             0.9842025637626648,
43             -0.008108563721179962,
44             -0.004064855165779591,
45             0.002380185993388295,
46             0.9450775980949402,
47             -0.07938937842845917,
48             74.02395629882812
49         ],
50         "std": [
51             0.09088966995477676,
52             0.07300057262182236,
53             0.1372314691543579,
54             0.028434785082936287,
55             0.15344616770744324,
56             0.0831201896071434,
57             0.15708106756210327,
58             0.09653346240520477,
59             0.25792863965034485,
60             17.60527801513672
61         ],
62         "q01": [
63             -0.26164972188472746,
64             -0.1068384879797697,
65             -0.34838534088134765,
66             0.8578828950166703,
67             -0.39593087464571,
68             -0.24848717963695527,
```

```

69         -0.4440641324043274,
70         0.5524548681385816,
71         -0.7954352568626404,
72         3.8117211180329327
73     ],
74     "q99": [
75         0.24469801027774807,
76         0.2604188863694668,
77         0.32515650482177727,
78         0.9999247654318809,
79         0.44039196008443837,
80         0.21469024567604067,
81         0.4175887481331826,
82         0.9998012677034364,
83         0.6080406239509584,
84         87.81824750663043
85     ]
86 }
87 }
88 }

```

代码块

```

1  {
2      "norm_stats": {
3          "state": {
4              "mean": [
5                  0.9866386651992798,
6                  0.01174874696880579,
7                  0.0863250195980072,
8                  -0.014884665608406067,
9                  0.9902113080024719,
10                 0.02754073217511177
11             ],
12             "std": [
13                 0.016135459765791893,
14                 0.11576970666646957,
15                 0.07273980230093002,
16                 0.11561324447393417,
17                 0.013251236639916897,
18                 0.07042448967695236
19             ],
20             "q01": [
21                 0.9218441331386567,
22                 -0.24597093497514724,
23                 -0.04534276025593281,

```

```
24         -0.3683149506390095,
25         0.9271771247386933,
26         -0.14952069258689882
27     ],
28     "q99": [
29         0.9999728871583938,
30         0.36374183725118636,
31         0.2584906660377979,
32         0.2395101139605046,
33         0.9999803764820099,
34         0.1763893915861845
35     ]
36 },
37 "actions": {
38     "mean": [
39         0.002787106204777956,
40         0.008149577304720879,
41         -0.01218277309089899,
42         0.9934353232383728,
43         0.0025930344127118587,
44         -0.000488636433146894,
45         -0.0045791370794177055,
46         0.9936944842338562,
47         -0.016093658283352852,
48         69.27120208740234
49     ],
50     "std": [
51         0.0603068470954895,
52         0.07054008543491364,
53         0.10658767074346542,
54         0.010426834225654602,
55         0.0840914398431778,
56         0.07680719345808029,
57         0.08486998081207275,
58         0.010469620116055012,
59         0.07055883854627609,
60         19.853910446166992
61     ],
62     "q01": [
63         -0.21753077259659767,
64         -0.1829988469451666,
65         -0.24897649617791176,
66         0.9486267318725586,
67         -0.20521489329338075,
68         -0.19385018956363204,
69         -0.3074472845435142,
70         0.9486010955810547,
```

```
71         -0.20536727986335754,  
72         -0.312166441488266  
73     ],  
74     "q99": [  
75         0.1901125014960766,  
76         0.17970724571049213,  
77         0.2357805479288101,  
78         0.9999780705451965,  
79         0.30566836494207383,  
80         0.16622908463180064,  
81         0.2085887821555138,  
82         0.9999774368286133,  
83         0.13457112607955934,  
84         87.07821368037462  
85     ]  
86 }  
87 }  
88 }
```

loss都是在归一化空间计算的，那为什么offline效果这么差，loss会下降到一个正常值？

2.26

xv_policy和umi_hybrid_policy的区别

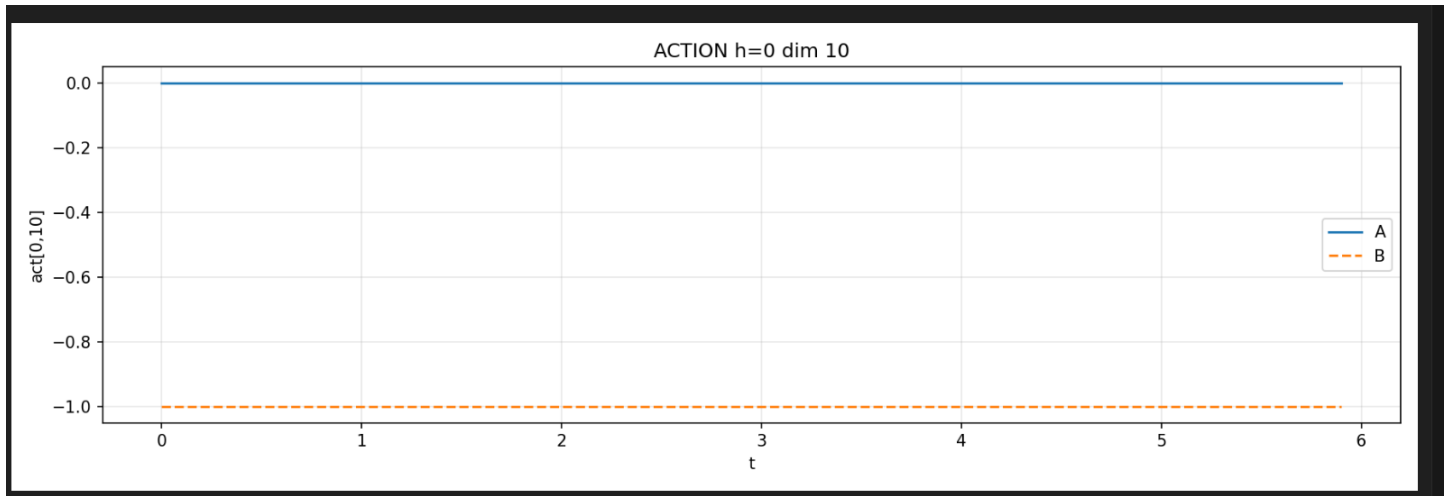
鬼故事2.0

不要显式加padding，让openpi自己加

显式加入0 padding后，再过归一化会被拉到 -1

A：未显式padding

B：显式padding



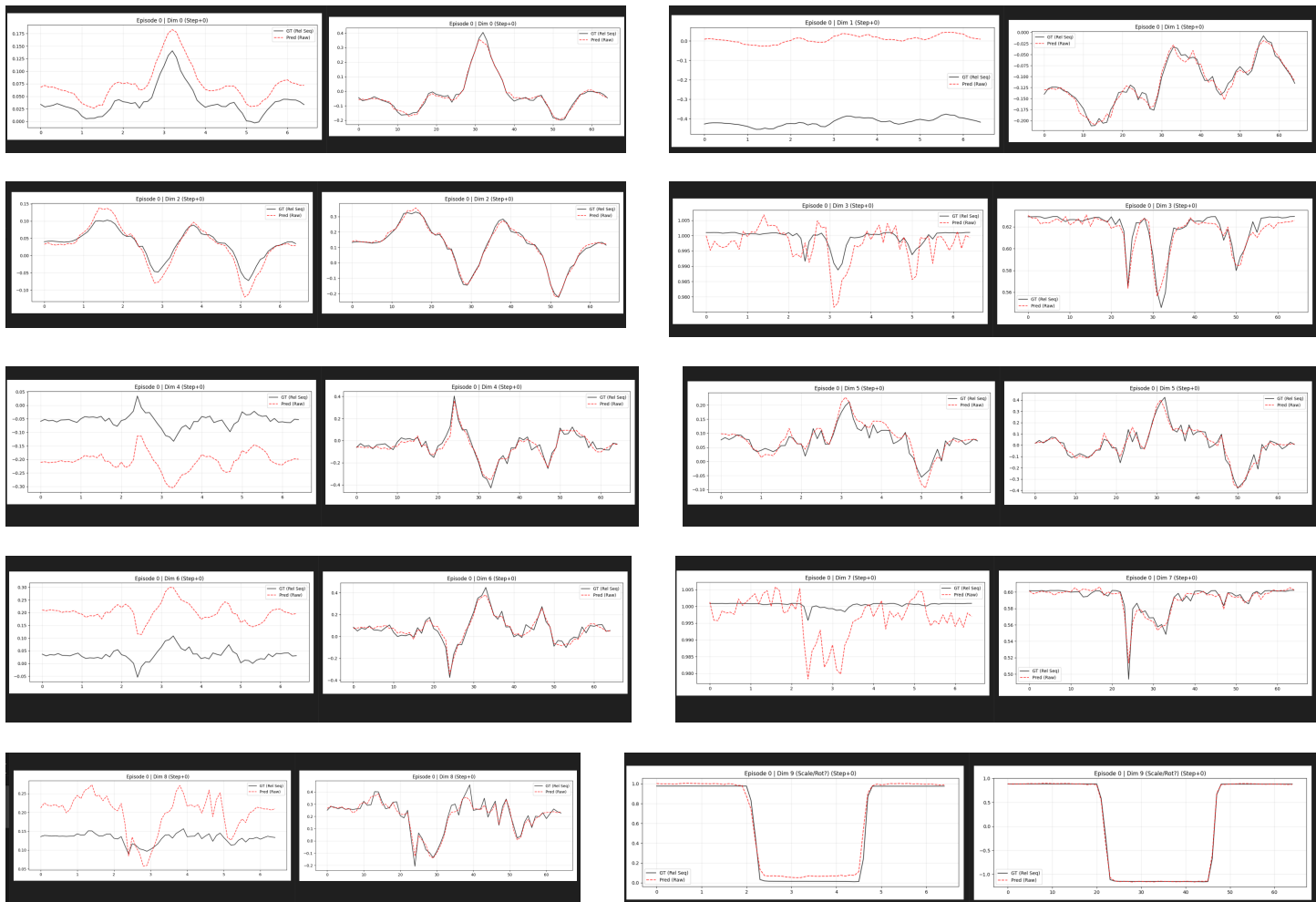
2.28

改用z zeros归一化

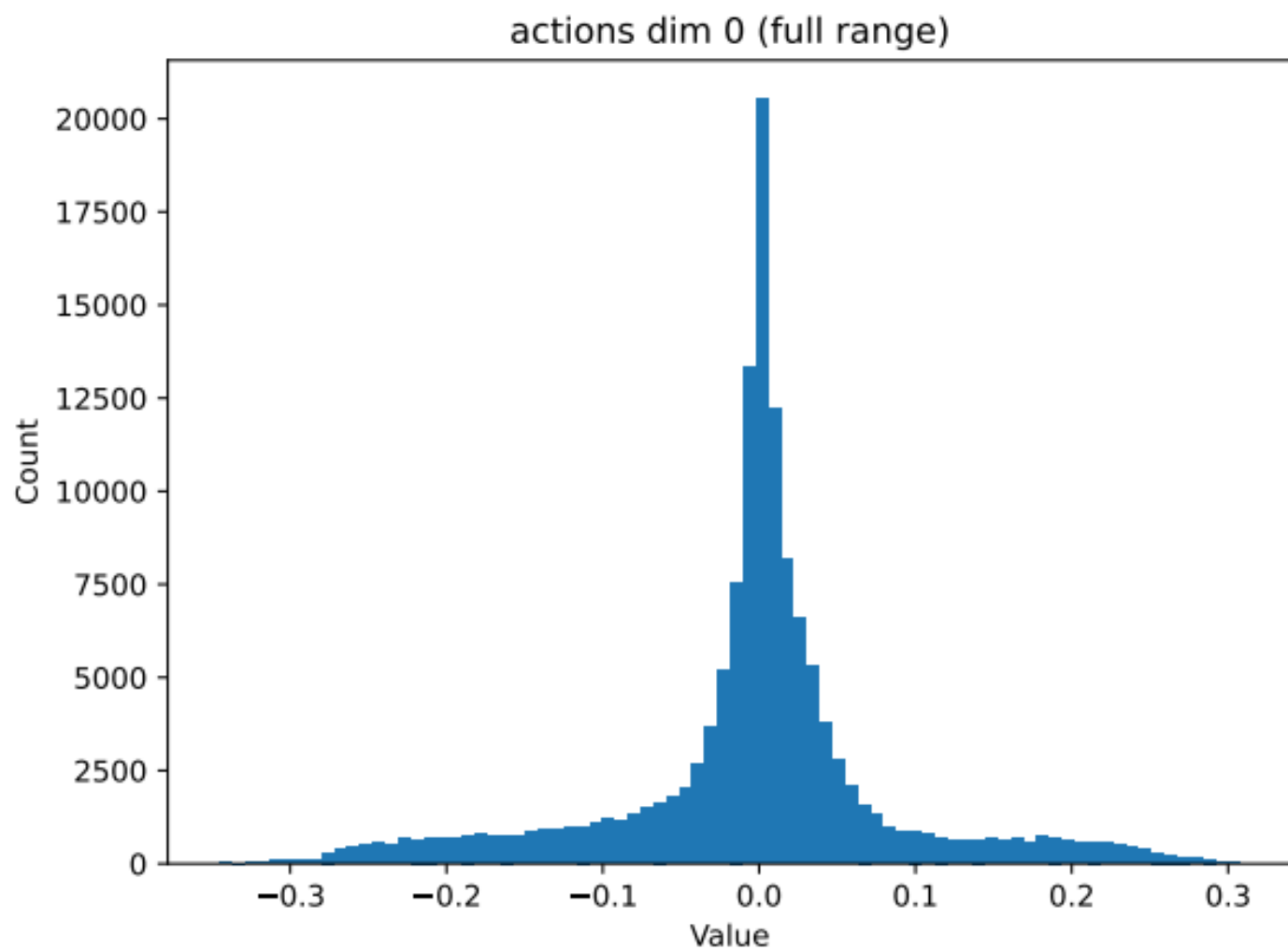
0228_s1: 好使数据 + z zeros

0228_s2: 不好使数据 + z zeros

改用z zeros归一化后offline问题解决，左侧为quantile归一化后offline情况，右侧为z zeros

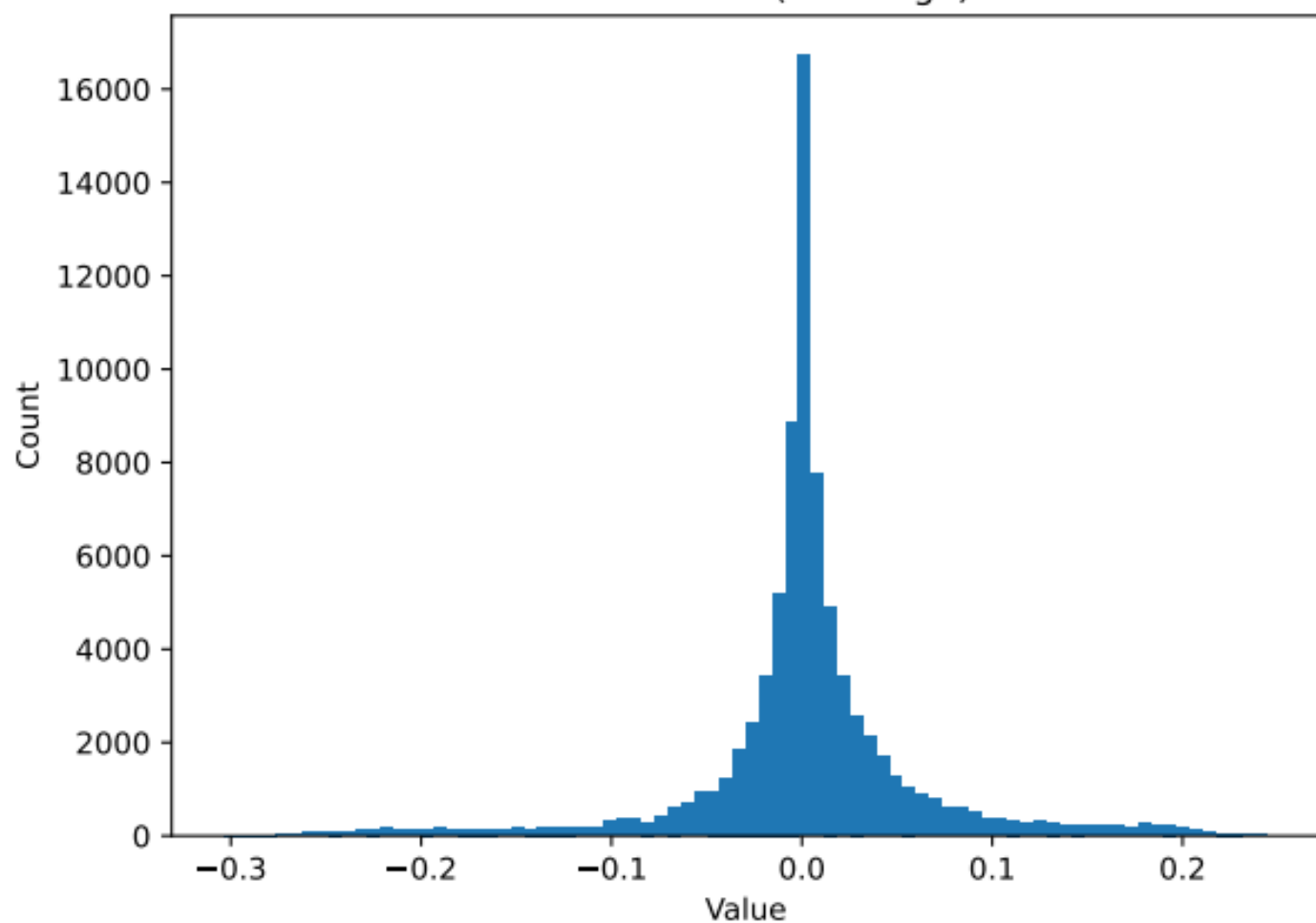


action和state的数据分布情况：

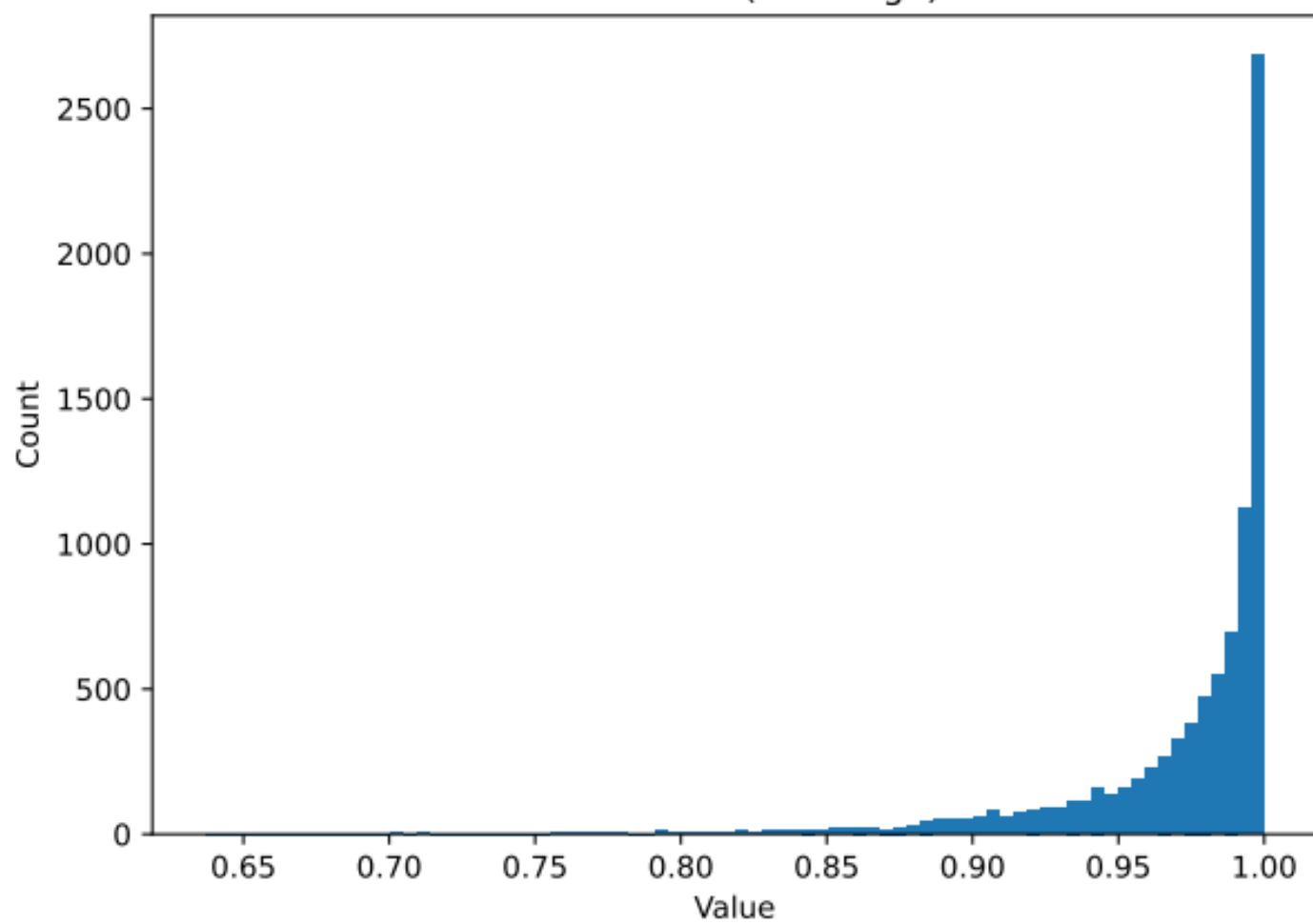


 actions_fullrange_histograms_csw.pdf

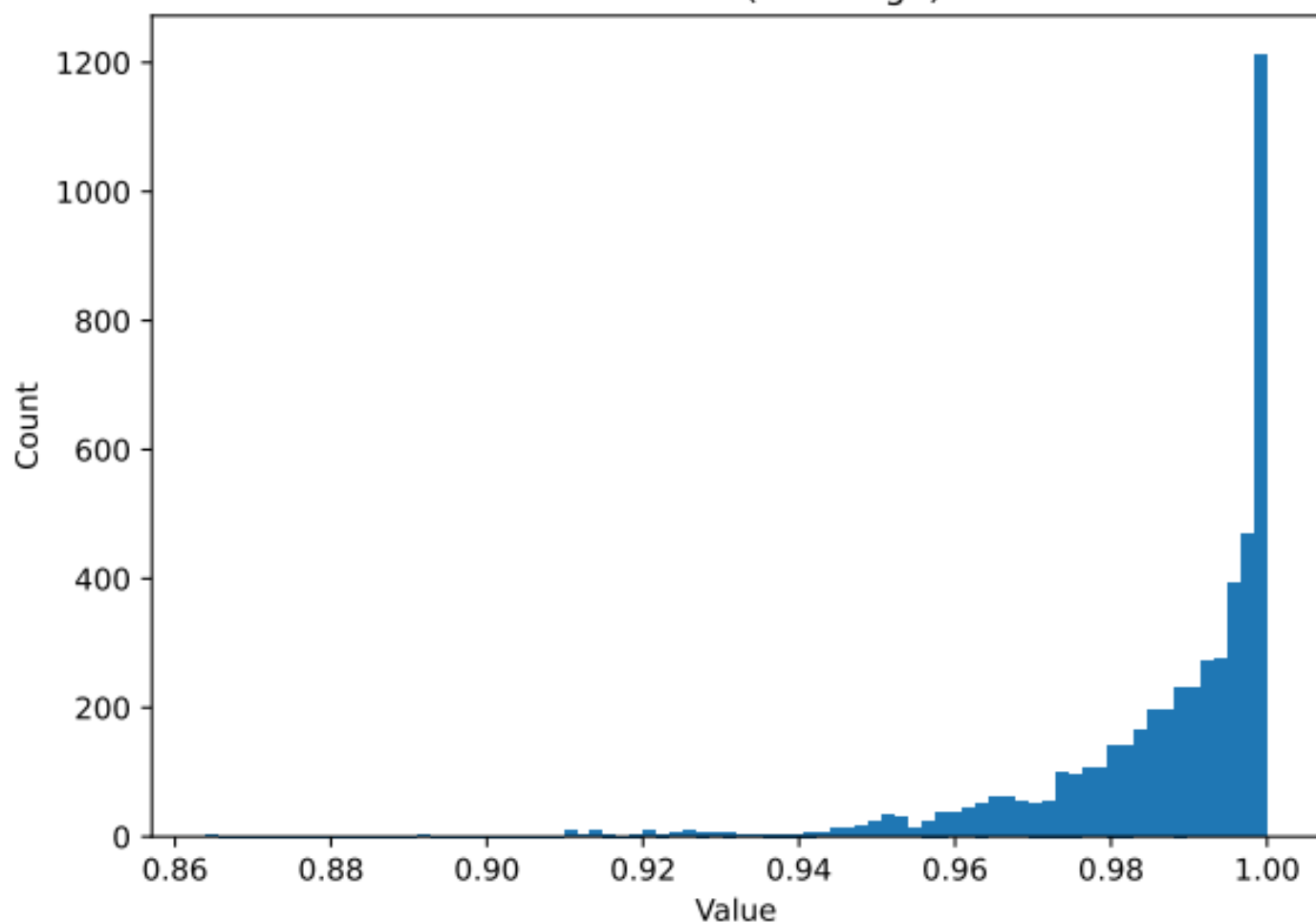
actions dim 0 (full range)



state dim 0 (full range)



state dim 0 (full range)



 state_fullrange_histograms_long.pdf

双臂操作

TODO

1. 场地setup

- 木质桌面
- 第三视角相机
- 另一个franka setup (robotiq+机载相机)
- 两个franka位置调整

2. 数据采集

- 笔记本扩容：内存&硬盘  单小龙
- 加扩展坞，支持双臂数据的采集，看下鹿明的数据采集平台  赵大为
- 加第三视角（桌面加固定的D435i）  陈帅文

- d. 加头部视角（3月份鹿明头箍相机到位）
- e. 支持遥操采集双臂，为后续部署提供基础能力 秋怡 顾秋怡
- f. 数据分析：为什么有些数据训不出来 ... 陈帅文 ， done
 - i. 改变归一化为z zeros即可，数据分布已做出，但是暂时找不到灵感
- g. Task
 - i. 叠纸盒
 - ii. handover 矿泉水
 - iii. 烤箱加热食物
- 3. 数据处理：需要转到同一个坐标系下
 - a. 用aruco marker将两个gripper的pose转到aruco坐标系 秋怡 顾秋怡
 - b. lerobot格式支持双臂
- 4. 模型训练
 - a. Dual arm policy image(wrist left +right) + state + action
- 5. 部署
 - a. 在一台4090上推理
 - b.

主从模式遥操采集

[📺 双臂Franka+第三视角](#)

[📺 gello双臂franka+第三视角](#)

无本体双臂

[📺 FastUMI Pro安装使用 双臂部分](#)

3.2

能采下来第一条数据



飞书20260302-161123.mp4

207.69MB



飞书20260302-161131.mp4

167.15MB



几个问题：

1. 数采平台要求每次初始位置固定，需要一个架子
2. 容易发生掉帧问题

3.3

无本体双臂各自是各自的base。。。

3.4

直接录制rosvbag和第三视角的情况，筛选完选了三条数据，看看能不能把后面的处理跑通，



挂了训练 0304

0304_34

35条数据，第三视角 + 双臂

3.22

双臂franka gello 遥操采集基本调通

四组数据：handover的循环任务，high代表一个较高的第三视角相机，各84条。不带high53条
采集训练文档随后整理

需要训：

handover任务

1. 高第三视角+lumos相机 0322_s1
2. 高第三视角+realsense相机 0322_s2
3. mask第三视角+lumos相机 84条 0322_s3
4. mask第三视角+realsense相机 84条 0322_s4
5. mask第三视角+lumos相机 84条+53条
6. mask第三视角+realsense相机 84条+53条
7. 混高低第三视角+lumos相机
8. 混高低第三视角+realsense相机

reverse_handover任务:

1. 高第三视角+lumos相机 0322_s5
2. 高第三视角+realsense相机 0322_s6
3. mask第三视角+lumos相机 84条 0322_s7
4. mask第三视角+realsense相机 84条 0322_s8
5. mask第三视角+lumos相机 84条+53条
6. mask第三视角+realsense相机 84条+53条
7. 混高低第三视角+lumos相机
8. 混高低第三视角+realsense相机

 **handover.zip**
276.59MB 

 **handover_high.zip**
415.40MB 

 **reverse_handover.zip**
262.38MB 

 **reverse_handover_high.zip**
381.95MB 

3.25

handover循环任务混训, handover_mix_high

`mix_high_lumos` (lumos) 和 `mix_high_rs` (rs)

原先mask逻辑有点问题, 重新训练0322的几个mask部分

0322_s1效果很差, 似乎没学会, 数据replay是没问题的, 感觉是训练过程出现bug

3.27

绝对位姿训练一个setting high lumos + mask lumos

`absdelta_high_lumos` `absdelta_mask_lumos`

check相对位姿训练模式，感觉没问题。。。。重新训几个setting看一下吧。。选择handover任务，high lumos 和 mask lumos rel_high_lumos mask_high_lumos

3.28-3.29

为franka悼念吧。。。。右臂getstate坏掉了，等修中解决

使用abs算delta的数据有问题，mean数值太小，导致推理结果很小，部署效果很差解决

1. 部署部分应该没问题，是训练的模型有问题
2. loss现在单卡 3w step loss 0.08 ，还是得开多卡，大bs到0.00x
3. 带第三视角的效果更差了，感觉是因为第三视角位置变化和环境变化

重新采数据，handover_0329,reverse_handover_0329

50几条



eva训练：handover0329_lumos_v1, bs=128

eva训练：reverse_handover0329_lumos_v1, bs=128

无问训练：handover0329_lumos_mask_v1, bs=64

无问训练：handover0329_rs_v1

4.1（遥操部分work）

补充部分数据，重点在对齐部分

考虑所有数据混到一起训练，两个任务也混合到一起，加起来有400左右数据

训练计划：

带第三视角 + lumos 八卡训练，无问 handover_third_lumos_400

mask第三视角 + lumos 四卡训练，无问 handover_mask_lumos_400

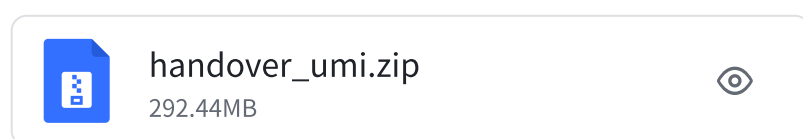
带第三视角 + D435 四卡训练，无问 handover_third_rs_400

mask第三视角 + D435 四卡训练，无问 handover_mask_rs_400

带第三视角 D435没训出来，其他效果如文档所示

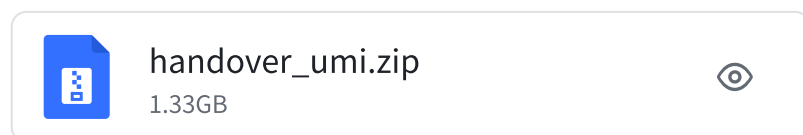
4.6

无本体采集handover数据，SLAM问题比较大



4.10

handover 200条 handover_umi_0410



fold_towel 60条 fold_towel_umi_0410



两个模型效果不好，action输出很小，机械臂基本不动

4.14

调大num_worker，现在调到16，cpu利用率上升，训练速度上去了

修改action_sequence_keys，和xvinput使用的key对齐

handover_umi_0414

效果不是很好，抓取部分不是很能抓到，递交过程会提前松手导致物品掉落

采数据还是得老老实实采，按照机器人的手法来做

fold_towel_umi_0414

4.15

加入一些random位置到操作物品的数据，缓解初始观测导致OOD的情况

random_approach 22条，加入脚踏板帮忙采数据

危险可能在于这些数据分布与原先handover差距很大，导致norm计算出现问题

handover_umi_0415

效果不好

4.16

加入脚踏板采集数据后，能够有一个还差不多的初始位置来采集，重新采数据做训练，31条

handover_umi_0416

效果不好

handover_umi_0416_mix 混合了之前的和random_approach

效果强于其他几个setting

4.17

关于双臂state的考虑：

使用二维码出第一帧的双臂位姿，用一个广角的带深度相机

可以得到初始双臂相对于相机base坐标系的位置，后面可以来一个世界坐标，算右臂在左臂坐标下的relative state作为双臂的输入

问题：

1. 第一帧两个夹爪必须出现在视野中
2. 需要一个广角的rgbd
3. 离线对数据进行后处理

4.20

handover_umi_0416_mix 效果不是很好，在第一阶段抓住物品后很容易松手，怀疑是数据比较脏，考虑对数据进行一个筛选，去掉特别脏的内容

handover_umi_0420_mix

效果好了一点,感觉（1）阶段的不那么容易OOD了，（2）对齐阶段表现也好了一点，感觉还是需要c这种，初始位置差不多的，然后采的慢一点的数据

4.21

30条良好数据，有二维码

4.22

handover_umi_0422_good_mix：好的数据混合

handover_umi_0422_mix：全部混合

handover_umi_0422_aruco：带二维码的数据，转换为relative的state

dev-qiuyi：稳定的双臂-第三视角训练pipeline

dev-csw：加入relative state+头部视角

4.23

完善数据清理和合并方案，用一些rule based的方案进行简单筛选，最终会有这样一个界面，同时完成筛选



 trajectory_anomaly_visual.html

采集200多条数据，分原始数据和exchange左右手

handover_umi_0422 210条 有一些nano录制内容，高质量数据，带二维码

handover_umi_0422_exchange 更换左右手，减少背景的不统一。 训练

handover_umi_0423_good_mix 250条 训练

0422_good_mix和handover_umi_0422 混合

handover_umi_0423_mix 将近500条，分高质量和低质量 训练

0422_mix和handover_umi_0422 混合，考虑加入质量分级

4.24

handover_umi_0422数据质量问题比想象的大一点，选出其中好的部分进行训练

handover_umi_0424 126条

handover_umi_0424_mix: 0422_good_mix和handover_umi_0424混合，算是优质数据混合，160条

4.29

handover_umi_0429

129条数据，有nano的记录

所有数据的混合：handover_umi_0429_mix。差不多600条

使用nano作为第三视角：handover_umi_0429_nano。 120条

